GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

Abstract. We study the finite positive memory required to realize causal experiments and implement their optimal policies. Multiple preparations sharing one controller and decoder admit a common permutation normal form under finite-group convex specifications. An invariant-polytope formula counts mixed memory and every prescribed pure initializer exactly. Joint Fourier multiplicities give the exact small-contrast core resource for preparations with a common group average, and shared memory can grow arbitrarily while each preparation alone needs two labels. For finite reset instruments with positive failure effects, positive costs and no closed failure set, a transversal formula classifies memory for robust optimal control with mandatory success stopping and tied actions. Attained variational certificates detect every insufficient memory, and explicit rational instruments realize arbitrary action-cover problems. Finally, a robust nonnegative-factorization argument gives memory lower bounds for arbitrary classical terminal decoders. The original algebraic qubit instrument has unbounded approximation memory with an explicit logarithmic-in-logarithm lower rate, without a badly-approximable premise or a Born representation of hidden labels. The general causal comparison and the complete reset, discrimination, variational, dual, spectral and geometric developments retain their individual hypotheses and full proofs.

1. Introduction

A predictive state records the information in an experimental past that can affect future laws. Its linear dimension measures one form of compression. A device storing a probability distribution on finitely many labels faces additional constraints: its transition and readout tables must be positive, its initialization must be implemented, and a single device may have to serve several preparations and later requests. These constraints are also present when the aim is to implement an optimal experimental policy rather than to reproduce a prescribed law. The resulting resource can be much larger than the predictive dimension or the resource of any one preparation considered separately.

The question of this paper is how predictive, convex and spectral structure controls such finite positive implementations. We work within a theory of prepared causal experiments, acquired task laws and finite-resource statistical decisions. A terminal response assigns a probability row to each command word. An instrument response describes joint command–report histories and the state on which a subsequent terminal test acts. All controllers have one fixed parameter-independent program. Persistent labels, policy storage, physical state, program bits and arithmetic work are different resources. An initializer is counted when it is a pure label of the device; freely prepared mixed starts are always stated as a separate convention.

Three structural results organize the current article. First, we classify the common positive realization of multiple preparations under finite-group response constraints. The exact initialization term is geometric, and the joint rather than individual Fourier multiplicity governs small-contrast memory. Second, we identify the exact number of fixed command rows needed for robust optimal control of finite reset experiments with tied actions. This number is a finite transversal invariant

Date: October 2026.

2020 *Mathematics Subject Classification.* Primary 60J10, 68Q45; Secondary 20C15, 52A20, 60G40, 62C20, 81P15, 90C22, 94A29.

Key words and phrases. causal experiment, predictive state, positive realization, minimal memory, finite group, quantum instrument, convex duality.

coupled to the optimal initial row. Third, we give finite operational memory certificates which do not require physical density matrices or Born readouts at hidden simulator labels. The original algebraic qubit experiment then has a provably diverging approximate memory requirement even in this larger decoder class.

One controller for several preparations. Fix a finite group G, a finite command alphabet whose words map onto it, and a finite family of preparation responses $b_i : G \to \Delta(C)$. Every preparation uses the same command matrices and terminal decoder. The decoder does not receive the preparation identity as an extra input. For closed convex specifications $K_{i,g}$, imposed on every word with image g, Theorem 5.1 gives a common permutation action P_g on a finite set X, one decoder D, and preparation rows λ_i satisfying

$$\lambda_i P_g D \in K_{i,g}.$$

Its pure-start label charge is exactly $|X| + \iota(\Lambda)$, where $\iota(\Lambda)$ counts the distinct nonpoint rows among the λ_i. In the freely mixed convention it is $|X|$. The original stochastic controller need not satisfy the group relations, and the resulting action need not be faithful.

For exact targets let $\mathcal{B}$ be the set of distinct prescribed response tables. Consider all G-invariant polytopes P in the product simplex of probability-valued functions on G that contain $\mathcal{B}$. With $v(P)$ the number of vertices, Theorem 5.2 proves

$$\kappa_G(\mathcal{B}) = \min_P v(P), \qquad m_G(\mathcal{B}) = \min_P \{v(P) + |\mathcal{B} \setminus \operatorname{vert} P|\}. \tag{1.1}$$

The minimum polytope is allowed to extend beyond the target linear span. Its vertices are continuing permutation states. Every distinct prescribed nonvertex response has one counted transient initializer. The proof also accounts for nonextreme continuing labels in an unreduced controller, so the formula does not charge such labels twice.

Let $V_{\mathcal{B}}$ be the real span of all translated preparation responses. If $A_\rho(b_i)$ is the stacked Fourier block for an irreducible representation ρ, its multiplicity in the joint response module is

$$r_\rho(\mathcal{B}) = \operatorname{rank}[\, A_\rho(b_1) \ \cdots \ A_\rho(b_{|I|}) \,].$$

This rank can exceed the degree of ρ, a possibility excluded by a single cyclic preparation. Lemma 5.3 and Theorem 5.4 show that when all preparations have a common group average π, the exact small-positive- contrast mixed memory of $\pi + t(b_i - \pi)$ is the minimum permutation degree containing this joint module. It is an integer cover by the fixed-space multiplicities of subgroups. A common positive decoder is constructed explicitly. The common-mean assumption is substantive: a four-preparation trivial-group example has linear rank three and positive memory four.

Theorem 5.5 replaces exact joint ranks by singular values above the specified approximation error, giving robust resource certificates. Two exact families expose the separate costs. With q preparations on the fixed group C_2, each preparation alone needs two labels but a shared controller needs $2q$; the obstruction persists at a positive explicitly bounded error. A second family has shared mixed memory two and pure memory $r + 1$ for r preparations. These are growing finite experiments, with arbitrary finite preparation count, rather than a change of numerical prior in one fixed example.

The inherited finite-group convex-specification theorem and its attained all-word minimax principle remain in the article with their full proofs. The supplement retains the exact one-preparation normal form, the complete mean–contrast law for the displayed family on every S_n, $n \geq 5$, its boundary rigidity, normal-stabilizer convexity and quotient duals. The shared-device results extend those premises without replacing a general positive-realization question by linear dimension alone.

Optimal feedback with ties and a counted initial row. The finite reset instrument has failure branches $\mathcal{I}_{a,i}(\rho) = \operatorname{tr}(E_{a,i}\rho)\sigma_i$ and a success branch on which the experiment must terminate. Its observed report prepares a specified state σ_i; no external reset or state oracle is supplied. Commands are arbitrary in a finite set, call costs have a positive floor, and the initial density matrix is unknown.

The inherited primitive theorem reduces its unrestricted optimum to a finite Bellman vector v and the spectral minimization

$$V_* = \min_{p\in\Delta(\mathcal{A})} \lambda_{\max}\Big(\sum_a p_a L_a\Big), \qquad L_a = G_a + \sum_i v_i E_{a,i}.$$

Its preparation saddle supplies a residual identity, but did not itself classify the exact number of rows of every optimal tied-action controller.

Assume now that every failure effect is positive definite and there is no closed failure set. Every reset report then occurs with positive probability on the first call under a single least-favorable preparation, including a singular such preparation. Let $\mathcal{A}_i^*$ be the set of Bellman-optimal actions at reset state i, $\mathcal{P}_*$ the set of optimal initial rows, and $I(p) = \{i : \operatorname{supp} p \subseteq \mathcal{A}_i^*\}$. If $t(J)$ denotes the least number of actions meeting every $\mathcal{A}_i^*$ for $i \in J$, then Theorem 7.1 proves

$$m_* = 1 + \min_{p\in\mathcal{P}_*} t(\{1,\dots,s\} \setminus I(p)). \tag{1.2}$$

There is no uniqueness assumption on optimal actions. Arbitrary stochastic transitions and mixed continuing rows cannot improve this number. An optimal implementation uses the chosen initial row and pure rows for a minimum transversal, updating solely from the reset report. Theorem 7.2 gives an attained finite variational cost lower which is strictly positive above V_* exactly at the insufficient label budgets.

The cover is an intrinsic obstruction within physical experiments. For every nonempty finite set system of proper action subsets, Theorem 7.3 constructs rational reset data with commuting cost and effect operators with constant success probability and bounded costs whose optimal memory is its transversal number plus one. The unrestricted value and unique uniform initial row are explicit. Thus the optimal memory can encode NP-hard covering instances even when the unrestricted optimal value is simple. This is a specific reduction; it is not an assertion that finite-state controller complexity is new in general. An exact two-dimensional counterexample shows why semidefinite failure effects alone do not support (1.2).

Approximation without a Born model of hidden labels. The earlier physical-dictionary construction gives logarithmic memory for the one-angle stopped qubit experiment, and a matching leading constant under a badly-approximable phase assumption. Its converse used a density matrix and Born terminal decoder at each hidden label. An arbitrary classical simulator need not have this representation.

Suppose J causally prepared heralds have target probabilities at least $p_* > 0$. After herald n, a terminal test m is requested. Its rejection probability is zero at $m = n$ and at least $s > 0$ otherwise. The request is delayed until after the preparation, and the terminal decoder receives no uncounted copy of the earlier history. Theorem 8.2 shows that a simulator approximating these joint records within total variation δ must have at least

$$a(J) = \min\{m : \tbinom{m}{\lfloor m/2\rfloor} \ge J\}$$

labels whenever $\delta \le p_* s^2/(32J^2)$. The certificate uses only a finite family of preparation–test experiments. It starts from joint herald-and-rejection events, controls herald mass, and only then divides to obtain a conditional factorization. A robust support-antichain argument permits completely arbitrary stochastic decoder columns.

For the original undamped qubit instrument with Bloch rotation $z = (3+4i)/5$, Theorem 8.3 gives the explicit finite certificate

$$\delta \le \frac{25\zeta}{128\cdot 200^J J^2} \quad\Longrightarrow\quad N_{\mathrm{cl}}(\delta; B) \ge a(J), \qquad \zeta = \min\{1, 7B/8\}. \tag{1.3}$$

The proof uses rare reset–rotation prefixes, delayed terminal projectors and an integer separation bound for powers of z. It needs no badly-approximable assertion about the angle. At fixed $B > 0$ it

yields a lower order $\log\log(1/\delta)$ for either start convention. An elementary algebraic norm argument extends the same order to every non-root algebraic phase on the unit circle. The inherited physical construction is admissible in this larger class and has order $\log(1/\delta)$; these bounds are not claimed to match. The sharp antichain example concerns a preparation–test table, not the qubit's entire chronological dynamics.

Classical ingredients and retained mathematical structure. Residual equivalence, probabilistic automata and positive realization are classical [44, 79, 48, 49, 54]. So are stochastic group structure, representation theory and equivariant factorization [61, 62, 75, 67]. The shared initialization invariant and noncyclic joint resource are the additional statements proved here. Bellman and stochastic-shortest-path theory [81], finite stochastic controller search [82, 83], and finite covering complexity [84] are used with their established meanings. Nonnegative rank and antichain lower bounds have explicit antecedents [67, 85]; the new operational claim includes the perturbation margin, stopped-cost accounting and algebraic-qubit application. A targeted comparison does not certify exhaustive priority.

The general actual-history comparison and its common paid construction remain in the main article. Its lower witness is evaluated on the competitor's actual reached histories. The main also retains the general stationary-initialization identity, separately from the new shared-preparation charge. The supplement preserves all earlier finite-order quantum measurements, two-stage reset, memory and communication, rank-constrained discrimination, performed-law compactness, variational and dual attainment, optimal strategy implementation, spectral memory and acquired-geometry arguments. The independent R58 companion remains separately readable and buildable. Their scopes are not replaced by a single group example or by the new positive-effect reset class. In particular, no result about response-memory complexity alone solves the original general-M squared-sign minimax-risk frontier or the independent A1/A2 programme.

2. Prepared kernels, acquired task laws and observers

All hidden-state, action and report spaces are standard Borel. A fixed unknown index $\theta \in \Theta$ specifies an actual preparation and positive normalized causal kernels. Positivity means nonnegative measures, not an everywhere strictly positive density. Dependence on the entire physical past may be included in the hidden state. A legal policy sees past visible reports and its admitted randomization, but not θ or a future audit. Iteration of these kernels defines its finite-prefix laws; the consistent countable path law is their kernel extension.

Each call has a cost $c_i \geq c_* > 0$. The decision to make the next call is predictable. In the geometric route, the declared productive-call flag is also predictable; in the primitive-instrument route, productive reports are specified outcomes and can be random. Commitment is permitted only after a productive report, and occurs exactly once almost surely in each model. At that time a specified future audit $Z \in H$ is scored, where $H \subset \mathbb{R}^k$ is nonempty, compact and convex. The prediction is committed before that audit is observed. A prospective joint report–audit kernel is defined at every productive call, including rejected calls. The stopping coin is independent of the audit conditional on the entire current visible report and the past. Write $L = \operatorname{diam}(H)^2$; the case $L = 0$ is trivial whenever a legal rule exists.

The core cost C_τ includes initial preparation charged to the core, failed calls and all productive calls through commitment. A separately executed final audit must be added to the raw budget. This convention is fixed before taking a risk infimum. Physical time, stored program constants and temporary computation are not identified with call count.

Three resource interfaces. A checkpoint procedure may use its whole precommit history, but must output one element of a fixed dictionary of at most M values. This counts its output width only; it is not a claim of $\log_2 M$ total working bits.

An autonomous observer has an alphabet S, $|S| \le M$, and one fixed parameter-independent program. At an intercall boundary its only persistent observer memory is its label $s \in S$. Its next command distribution is a fixed function of s. A physical port name, elapsed time, subprogram index, random seed or former report is not an extra free input. During a call its current command, current report and fresh coins are available. Whatever persists to another call must be encoded in S. A fixed decoder $r : S \to H$ reads only the committed label.

In the *event-stopped* interface a post-report kernel emits $(s', e) \in S \times \{0, 1\}$. An event $e = 1$ ends the experiment and retains s' for the decoder; no subsequent command is executed. This public commitment event is not a stored bit used in continuing computation. The same label may be used live on one path and terminal on another. In the *state-stopped* interface a fixed map $h_* : S \to \{0, 1\}$ determines halt status. The initial label is live, and type-one labels are globally absorbing terminal labels with no executed command rows. Entry into a terminal label is allowed only after a productive report. Thus live and terminal types are disjoint in this interface, and only in this interface by definition.

Write $R_i^{\Theta}(N, M)$, $i \in \{\mathrm{cp}, \mathrm{ev}, \mathrm{st}\}$, for the infimum of worst-case expected squared loss over programs with robust almost-sure commitment and $\sup_\theta \mathbb{E}_\theta C_\tau \le N$. An empty legal class has value $+\infty$. A program chosen once at random is not a free persistent seed. The ideal measurable class may have fixed real constants; effective implementation is stated separately in Section S41.

A physically acquired comparison law. Execute a declared prepared comparison experiment and obtain its joint report–audit probability $K_0(dx, dz)$ on $X \times H$. Its report marginal is ν. The law is genuinely preparable and its specification has its own resource account. It need not be a member of Θ. In particular no finite set of calibration data is silently promoted to a uniform certificate. Set

$$f(x) = \int z K_0(dz \mid x), \quad w(x) = \int \|z - f(x)\|^2 K_0(dz \mid x), \quad d_r(x) = w(x) + \|f(x) - r\|^2. \tag{2.1}$$

Disintegration gives Borel versions with $f \in H$; null sets may be completed arbitrarily. The profile is

$$\mathcal{Q}_m(b) = \min_{\substack{C \subset H,\ 1 \le |C| \le m \\ 0 \le h \le b,\ \int h\, d\nu = 1}} \int [w + \mathrm{dist}(f, C)^2] h \, d\nu, \qquad b \ge 1,\ m \ge 1. \tag{2.2}$$

We use $+\infty$ for an infeasible cap or width. The quantities f, w are task coordinates of acquired predictive information, not a complete substitute for the predictive quotient. Their derivation and the quotient construction in Appendix S42 precede any use of local dimension.

Lower certificates on actual histories. Fix *one* actual model $\theta_* \in \Theta$, independent of the competing program. At each actual reached productive pre-call history h and legal command a, write $P_{h,a}(dy, dz)$ for its conditional full current report–audit law. Let $P^0_{h,a}$ be a measurable comparison row on the same report space, with a Borel mark $x = \pi_{h,a}(y)$ of law ν and

$$\mathbb{E}^0[Z \mid y] = f(\pi_{h,a} y), \qquad \mathbb{E}^0[\|Z\|^2 \mid y] = w(\pi_{h,a} y) + \|f(\pi_{h,a} y)\|^2. \tag{2.3}$$

These are conditional identities given the *full* report, not just its mark. Versions and inequalities are required on actual reachable histories, not only almost everywhere under an unrelated comparison path law.

Fix $0 < l_- \le 1 \le u_- < \infty$. For all Borel $q : Y \to [0, 1]$ and all finite-image Borel $r : Y \to H$, suppose predictable nonnegative defects $a_-(h, a), e_-(h, a)$ obey

$$\int q \, d(P_{h,a})_Y \le u_- \int q \, d(P^0_{h,a})_Y + a_-(h, a), \tag{2.4}$$

$$\int q(y) \|z - r(y)\|^2 P_{h,a}(dy, dz) \ge l_- \int q(y) \|z - r(y)\|^2 P^0_{h,a}(dy, dz) - e_-(h, a). \tag{2.5}$$

Independent current randomization can be adjoined to both rows: integrating inequalities for each fixed random value proves the same inequalities with the same defects. All command-conditioned lower inequalities hold before averaging the chosen command.

Let A_τ count productive calls. At this same actual model require, for every legal competing procedure,

$$\mathbb{E}_* C_\tau \geq \kappa_- \mathbb{E}_* A_\tau, \qquad \kappa_- > 0. \tag{2.6}$$

At budget N suppose, uniformly over all robustly feasible competitors,

$$\mathbb{E}_* \sum_{j \leq \tau:\ j \text{ productive}} a_-(H_{j-1}, a_j) \leq A_N, \qquad \mathbb{E}_* \sum_{j \leq \tau:\ j \text{ productive}} e_-(H_{j-1}, a_j) \leq E_N. \tag{2.7}$$

These are actual occupation certificates, not properties merely of the proposed upper program. Uniform row defects give $A_N \leq (N/\kappa_-)\bar{a}_-$ and $E_N \leq (N/\kappa_-)\bar{e}_-$. Lemma S39.6 supplies a nonuniform alternative.

One common upper construction. Independently, let $\mathcal{G}$ be a parameter-independent program which, from each of its start histories, reaches one productive report. Its conditional expected attempt cost is at most $\kappa_+ > 0$ on every surviving history in every $\theta \in \Theta$. It uses B live roles and returns to its start label on rejection; no reset of the physical state is assumed. It forms a virtual mark x, a current choice J and a permitted export $T_J r$ of its task readout. For every finite dictionary C assume

$$|\{T_j r : j \in J,\ r \in C\}| \leq D_0 |C|, \qquad D_0 \in \mathbb{N}. \tag{2.8}$$

Every continuing phase and coin needed for this construction is included in the B live roles. For fixed $0 < l_+ \leq 1 \leq u_+ < \infty$ and $a_+, e_+ \geq 0$, assume for all Borel $q : X \to [0,1]$ and finite-image Borel $r : X \to H$, conditionally on every such history,

$$\mathbb{E}[q(x)] \geq l_+ \int q \, d\nu - a_+, \tag{2.9}$$

$$\mathbb{E}[q(x) \| Z - T_J r(x) \|^2] \leq u_+ \int q(x) d_{r(x)}(x) \nu(dx) + e_+. \tag{2.10}$$

The expectation here includes the within-attempt choice J. Unlike the lower, the upper is not required to hold for each choice separately. The choices retained after this call are nevertheless charged through the actual export or live role. These hypotheses express one common causal construction, not modelwise optimizers.

3. Reversible response groups and common controlled memory

The fixed-phase results concern one transition matrix. Several commands require one family of matrices on the same retained labels, and checking every command in isolation is insufficient. We give an intrinsic finite-realization criterion for responses whose minimal linear dynamics is reversible and compact, including any finite family of unitary commands on a prepared finite-dimensional quantum system. We then obtain a matching arithmetic memory law for a pair of noncommuting involutions. This is a theorem about exact controlled response realization, not an assertion that a simulator's private label is an observed sufficient statistic for arbitrary feedback optimization.

Positive realization through invariant polyhedral sets is classical [48, 47, 46]. The argument below uses a section of the realizing simplex before passing to the observable quotient; auxiliary hidden-state responses need not belong to the minimal response space. Compact reversibility then turns containment into equality. The resulting finite-group obstruction concerns all command words, rather than a single spectral restriction.

3.1. **The minimal response space.** Let A and C be nonempty finite alphabets of commands and terminal outcomes. A supplied response is a function $b : A^* \to \Delta(C)$. Its residual after a word w is $b^w(z) = b(wz)$, including the empty word $\varnothing$. Put

$$V_b = \operatorname{span}_{\mathbb{R}}\{b^w : w \in A^*\}, \qquad (L_a f)(z) = f(az).$$

Assume V_b is finite dimensional. It is invariant under each L_a, since $L_a b^w = b^{wa}$. Call the response *compactly reversible* if the restrictions $L_a|_{V_b}$ are invertible and the closure G_b of their generated group is a compact subgroup of $\mathrm{GL}(V_b)$. This concerns the observable quotient; it does not require the full physical command group to be finite or fully observable.

An exact stochastic realization with M live labels has row-stochastic matrices Q_a, one row-stochastic terminal decoder matrix D, and the prescribed initial row e_1^{T}, such that

$$b(a_1 \cdots a_n) = e_1^{\mathsf{T}} Q_{a_1} \cdots Q_{a_n} D. \tag{3.1}$$

Randomized terminal readout is allowed even when all Q_a are deterministic. The target input word is not encoded into an uncounted clock or persistent seed.

Theorem 3.1 (Finite realization of compactly reversible responses)**.** *For a compactly reversible response, the following are equivalent:*

(i) *An exact finite stochastic realization exists.*

(ii) *The residual set $\{b^w : w \in A^*\}$ is finite.*

(iii) *The observable group G_b is finite.*

(iv) *An exact realization with deterministic transitions and a possibly randomized decoder exists.*

The least number of deterministic-transition labels is the residual cardinality. This is not in general the least stochastic label count. If an M-label stochastic realization exists, then

$$|G_b| \le (2^M - 1)!.$$

This last bound is only a finite combinatorial certificate, not a sharp memory law.

Proof. Suppose (3.1) holds, and write $r_w = e_1^{\mathsf{T}} Q_w$. Let $R = \operatorname{span}_{\mathbb{R}}\{r_w\} \subseteq \mathbb{R}^M$. The assignment $\pi(r_w) = b^w$ extends to a well-defined surjective linear map $\pi : R \to V_b$: a relation $\sum_w c_w r_w = 0$ implies, after right multiplication by $Q_z D$, the identical relation among $b^w(z)$ for every z. Moreover

$$\pi(rQ_a) = L_a \pi(r) \qquad (r \in R).$$

The simplex section $K = R \cap \Delta_M$ is a compact polytope, invariant under right multiplication by every Q_a. Its image $P = \pi(K)$ is a compact polytope, contains every residual, and spans V_b. It satisfies $L_a P \subseteq P$. It is important that we intersect the simplex with R first: projecting arbitrary hidden-state responses directly into V_b would not be justified.

For an element T of a compact matrix group, positive powers have a subsequence tending to the identity with exponents tending to infinity. One elementary proof takes a convergent subsequence of powers with successively growing index gaps; quotients of two close powers approach the identity. A finite-order element is covered by its period multiples. Thus positive powers also approach T^{-1}. Applying this to L_a and using closedness of P gives $L_a^{-1} P \subseteq P$, and hence $L_a P = P$. All elements of G_b preserve P. They permute its finitely many vertices, and this permutation action is faithful because those vertices span V_b. Consequently G_b is finite. This proves (i)⇒(iii).

For the displayed bound, a vertex of K is determined by its positive support. If two distinct vertices had the same support, their difference could be added and subtracted in sufficiently small multiples while preserving R, unit mass and nonnegativity, contradicting extremality. Thus K has at most $2^M - 1$ vertices. A linear image of a polytope has no more vertices than the original polytope, so the faithful vertex action proves the bound.

If G_b is finite, the residual orbit is finite. Conversely, if the residual set is finite, each invertible L_a maps it injectively into itself, hence permutes it. Since it spans V_b, the generated group and its closure are finite. This proves (ii)⇔(iii). Under (ii), use the residuals as labels, send b^w to b^{wa}

on input a, and use decoder $b^w(\varnothing)$. This realizes b with deterministic transitions. If any such deterministic-transition machine sends u and v to the same label, every suffix yields the same decoder law, so $b^u = b^v$. This proves its minimal count. Finally (iv)⇒(i) is immediate. No assertion that stochastic transitions can always be determinized without increasing size was used. □

Corollary 3.2 (Prepared unitary commands)**.** *Let U_a be finitely many unitary matrices on $\mathbb{C}^d$, let ρ be a prepared density matrix, and let $\{E_r : r \in C\}$ be a terminal positive-operator-valued measurement. For*

$$b(w, r) = \operatorname{tr}(E_r \Phi_w(\rho)), \qquad \Phi_a(X) = U_a X U_a^*,$$

exact finite stochastic realization for all command words is possible if and only if its observable response group is finite. The group is computed after quotienting only those directions invisible to every future commanded terminal test. Its dimension is at most d^2.

Proof. Let $W = \operatorname{span}_{\mathbb{R}}\{\Phi_w(\rho)\}$ in the Hermitian matrices. Each conjugation is a Hilbert–Schmidt isometry. Its forward invariant subspace W is also backward invariant, by finite-dimensional invertibility. Let

$$N = \{X \in W : \operatorname{tr}(E_r \Phi_z X) = 0 \text{ for all } z, r\}.$$

It is invariant under every Φ_a and hence under its inverse. The orthogonal complement of N in W is invariant as well. Therefore the induced transformations on W/N are orthogonal for the quotient inner product. The map $X \mapsto (z, r \mapsto \operatorname{tr}(E_r \Phi_z X))$ identifies W/N with V_b and intertwines them with L_a. Their generated group has compact closure in the orthogonal group. Apply Theorem 3.1. A constant measurement can make this induced group trivial even when the physical unitary group is infinite, so finiteness of the full physical group is not a necessary condition. □

3.2. **A matching memory law for two noncommuting commands.** Fix $x \in \mathbb{R}$ with $\sin(2\pi x) \neq 0$. On a protected qubit take

$$U_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_1 = \begin{pmatrix} \cos(\pi x) & \sin(\pi x) \\ \sin(\pi x) & -\cos(\pi x) \end{pmatrix}, \quad \rho = E = |0\rangle\langle 0|.$$

Both commands are involutions. For known $0 \leq \alpha \leq 1$, their channels are

$$\Phi_{a,\alpha}(X) = \alpha U_a X U_a^* + (1 - \alpha) \operatorname{tr}(X) I/2.$$

Let $M_x^{\mathrm{ctrl}}(\alpha)$ be the least stochastic live memory of the whole binary response $w \mapsto \operatorname{tr}(E \Phi_{w,\alpha} \rho)$, using the same matrices and decoder for every word. In the real Bloch plane, with coordinates (z, x), these commands are αS_0 and αS_1, where S_0 reflects the angle t to $-t$ and S_1 reflects it to $2\pi x - t$. Their product is $\alpha^2 R_{2\pi x}$.

Theorem 3.3 (All-word arithmetic law for an involutive command pair)**.** *For $0 < \epsilon < 1$ and $\alpha = 1 - \epsilon$, the arithmetic envelope of Section S18 gives*

$$\tfrac{1}{4}\mathfrak{A}_x(\epsilon) \leq M_x^{\mathrm{ctrl}}(1 - \epsilon) \leq (1 + 4\pi^2)\mathfrak{A}_x(\epsilon). \tag{3.2}$$

The upper is one explicit finite stochastic machine valid for every mixed command word, not separately chosen unary machines. The lower ranges over arbitrary common stochastic realizations.

For irrational x, the lower and upper logarithmic growth exponents as $\epsilon \downarrow 0$ are respectively

$$\frac{1}{\mu(x) + 1}, \qquad \frac{\mu(x) - 1}{2\mu(x) - 1},$$

with values $0, 1/2$ at $\mu = \infty$. A uniform cubic-root law holds precisely for badly approximable x; otherwise the normalized count has lower limit zero and upper limit infinity.

At $\alpha = 1$, the exact count is the reduced denominator q if x is rational, and is infinite otherwise. At $\alpha = 0$, it is two. Every strictly damped pair has a finite exact common realization. These statements concern controlled response memory, not a general multi-command boundary-optimal policy theorem.

Proof. Any common realization restricted to the words $(01)^n$ realizes

$$\tfrac{1}{2}[1+\alpha^{2n}\cos(2\pi n x)]$$

with transition matrix Q_0Q_1. Its size is therefore at least $m_x(1-\alpha^2) \geq \mathfrak{A}_x(\epsilon(2-\epsilon))/2$ by Theorem S18.1. For $1 \leq c \leq 2$, the defining maximum of the envelope obeys $\mathfrak{A}_x(c\epsilon) \geq \mathfrak{A}_x(\epsilon)/c$ whenever both arguments are in its domain. This follows for each denominator before taking its infimum. Taking $c = 2-\epsilon$ proves the lower in (3.2).

For the upper, the rational multiple-polygon construction in the proof of Theorem S18.1 gives a regular l-gon P on the unit circle, with vertex $v_0 = (1,0)$, such that

$$l \leq (1+4\pi^2)\mathfrak{A}_x(\epsilon), \qquad \alpha R_{2\pi x} v_j \in P \quad \text{for every vertex } v_j.$$

The first reflection S_0 permutes this regular polygon and zero belongs to it. Consequently $\alpha S_0 P \subseteq P$, while $S_1 = R_{2\pi x}S_0$ gives $\alpha S_1 P \subseteq P$. Choose barycentric rows $Q_a(j,k)$ of $\alpha S_a v_j$ in the same vertices. Let $d_j = (1+(v_j)_1)/2$. Starting at vertex zero, induction on the actual command word shows that the expected retained vertex equals the protected Bloch vector. Pairing with the first coordinate proves every response identity. The start is one of the counted vertices; there is no extra initialization mixture.

The two constants in (3.2) transfer both exponents and both uniform-law statements from Theorems S18.3 and S18.4. At zero damping, a rational phase of reduced denominator $q \geq 3$ has a common deterministic q-gon realization. The word-pair transition has a primitive qth-root mode and needs at least q states by the unit-circle spectral obstruction. For irrational x that same obstruction rules out every finite size. At $\alpha = 0$ the response is one at the empty word and $1/2$ after every nonempty word, realized by a first label followed by an absorbing fair label; one label cannot change its decoder. This proves all endpoints. □

A fully algebraic example uses $U_1 = 5^{-1/2}\left(\begin{smallmatrix} 2 & 1 \\ 1 & -2 \end{smallmatrix}\right)$. Then $e^{2\pi i x} = (3+4i)/5$ is not a root of unity, since its sum with its conjugate is the rational noninteger $6/5$. Each undamped command in isolation has period at most two; their common all-word response has no finite stochastic realization. The arithmetic exponent of this particular angle is not determined here. The example isolates the failure of commandwise testing without inferring a Diophantine class from finitely many coefficients.

If an independent herald with fixed probability p is placed before each command operation, and only failures apply the supplied command, every common realization above preserves the terminal response under every external command schedule and the first-herald cost $1/p$. The commands are actual inputs to that same retained-state transducer. This does not make its private label observable to the caller, allow uncharged midstream quantum measurements, or prove optimality for a controller free to choose an easier command sequence. The unary optimal squared-task interpretation below continues to use its separately proved physical embedding.

4. Convex specifications and optimal uniform approximation

The exact normal form has a robust extension. The target need not be an exactly specified probability table, and the stochastic controller need not satisfy the group relations, even on its observable states. What is required is a constraint on every word representing a given group element. The quantifier over all words is essential.

Fix a finite command alphabet A, a finite group G, and a surjective homomorphism $\gamma : A^* \to G$. For every $g \in G$ let K_g be a closed convex subset of the finite probability simplex $\Delta(C)$. Empty sets are allowed. An M-label pure-start controller consists of a prescribed point row e_o, stochastic command matrices Q_a, and a row-stochastic decoder D_0. It satisfies the specification if

$$e_o Q_w D_0 \in K_{\gamma(w)} \qquad (w \in A^*). \tag{4.1}$$

For a mixed start, replace e_o by an arbitrary probability row and do not charge an additional label for preparing that row.

Theorem 4.1 (Finite-group convex-specification normal form)**.** *The specification* (4.1) *has a pure-start realization on at most M labels if and only if there are a finite G-set X, a probability row λ on X, and a row-stochastic decoder D such that*

$$\lambda P_g D \in K_g \quad (g \in G), \qquad |X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \le M. \tag{4.2}$$

For mixed starts the corresponding criterion is $|X| \le M$. The action on X is permitted to have a kernel. In particular no faithfulness hypothesis is imposed on an approximating action.

Proof. We give the projection argument explicitly because its use of convexity replaces the exact observable intertwining in Theorem S1.1. Let $\mathcal{L} = \gamma^{-1}(e)$. Select finitely many identity words $\ell_1, \dots, \ell_s$ whose positive matrix entries cover every directed edge occurring in any Q_ℓ, $\ell \in \mathcal{L}$, and put

$$T = (I + Q_{\ell_1} + \cdots + Q_{\ell_s})/(s+1).$$

At most M^2 words suffice. Each closed communicating class C_i of T is closed under every identity-word matrix. The positive diagonal and Lemma S16.1 give a stochastic idempotent $E = \lim_n T^n$. Its image simplex has disjointly supported vertices $\mu_1, \dots, \mu_k$, the stationary rows of the closed classes. Every identity word preserves those classes, and consequently

$$EQ_\ell E = E \qquad (\ell \in \mathcal{L}). \tag{4.3}$$

For a word u write $F_u = EQ_uE$ on this image simplex. Choose an inverse word $\bar{u}$ with $\gamma(\bar{u}) = \gamma(u)^{-1}$. Since T^n is a convex combination of identity-word matrices, (4.3) applied to $u\ell\bar{u}$ and $\bar{u}\ell u$ gives

$$F_u F_{\bar{u}} = E = F_{\bar{u}} F_u.$$

Thus F_u is an affine automorphism of the image simplex and permutes its vertices. If $\gamma(u) = \gamma(v)$, multiplication by the common inverse $F_{\bar{u}}$ shows $F_u = F_v$. Replacing the middle E by T^n in EQ_uEQ_vE also gives $F_uF_v = F_{uv}$, since all words $u\ell v$ have the same group image. This defines a G-action P_g on $X = \{1, \dots, k\}$.

Write $e_oE = \sum_i \lambda_i \mu_i$ and put $D(i, \cdot) = \mu_i D_0$. For any representative u of g,

$$\lambda P_g D = e_o E Q_u E D_0 = \lim_{n\to\infty} e_o T^n Q_u T^n D_0.$$

The last expression before passage to the limit is a convex combination of $e_oQ_{\ell u \ell'}D_0$ with $\ell, \ell' \in \mathcal{L}$. All these probability rows belong to K_g. Closedness and convexity therefore give $\lambda P_g D \in K_g$. This argument includes u empty and makes no claim that the original controller's word laws factor through G.

If the initial label lies in a closed class C_i, then $e_oE = \mu_i$ and λ is a point mass. Otherwise the k disjoint nonempty closed classes exclude at least the initial label, so $k + 1 \le M$. This proves the pure-start charge. For a mixed row η, use ηE instead and retain $k \le M$.

Conversely, a point preparation on X needs only the permutation transitions. For a mixed λ, a further label $\star$ has decoder λD and, on the first actual command a, moves into X with row $\lambda P_{\gamma(a)}$. Subsequent transitions are permutations. Its empty and nonempty word laws are respectively λD and $\lambda P_{\gamma(w)} D$. This gives the required realization and finishes both start conventions. □

For an exact probability table b, let $\kappa_G(b)$ be the least $|X|$ among factorizations $b(g) = \lambda P_g D$. This is its mixed-start minimum by the preceding theorem. The notation $m(b)$ retains the counted pure-start convention; its exact formula and the residual cardinality N are given in Theorem S1.2.

For a target $b : G \to \Delta(C)$ write $\mathrm{TV}(p, q) = \frac{1}{2}\sum_c |p_c - q_c|$ and define

$$\mathcal{E}_M^{\mathrm{p}}(b) = \inf_{\text{pure-start controllers on at most } M \text{ labels}} \sup_{w \in A^*} \mathrm{TV}(b(\gamma(w)), e_o Q_w D_0).$$

Use $\mathcal{E}_M^{\mathrm{m}}$ for the mixed-start convention. For a finite G-set X put

$$d_X(b) = \min_{\lambda, D} \max_{g \in G} \mathrm{TV}(b(g), \lambda P_g D), \qquad d_X^{\mathrm{pt}}(b) = \min_{x, D} \max_{g \in G} \mathrm{TV}(b(g), e_x P_g D).$$

The decoder ranges over stochastic rows; compactness makes both minima attained. Minima over no candidate actions are interpreted as $+\infty$.

Theorem 4.2 (Attained finite optimization for infinite-word error)**.** *For every target b, finite G, and integer $M \geq 1$,*

$$\mathcal{E}_M^{\mathrm{m}}(b) = \min_{|X| \leq M} d_X(b), \tag{4.4}$$

$$\mathcal{E}_M^{\mathrm{p}}(b) = \min\left\{ \min_{|X| \leq M} d_X^{\mathrm{pt}}(b), \min_{|X| \leq M-1} d_X(b) \right\}. \tag{4.5}$$

One representative of each finite G-set is sufficient, so these are finite optimizations and their minima are attained. Their values are 1-Lipschitz functions of b in $\max_g \mathrm{TV}$. If $\mathcal{E}_M^{\mathrm{p}}(b) > 0$, a sufficiently small perturbation of b still cannot be exactly realized on M pure-start labels; the same assertion holds for mixed starts. When the target table is algebraic, the optimum and a realizing finite controller are computable by real quantifier elimination. This is a decidability assertion, not a polynomial-time bound or a claim that quantifier elimination was executed.

Proof. Apply Theorem 4.1 to the closed convex balls $K_g = \{p : \mathrm{TV}(p, b(g)) \leq \delta\}$. It sends every controller of error at most δ to an admissible action without increasing the label charge or the error. Conversely every displayed action supplies such a controller. The pure constraint splits precisely into a point preparation with $|X| \leq M$ or an arbitrary preparation with $|X| \leq M - 1$. There are finitely many homomorphisms from the fixed finite group into each symmetric group of degree at most M. For each action the allowed preparations and decoders form compact sets. Hence the finite minima are attained, including limits in which a mixed preparation becomes a point mass. The triangle inequality proves the Lipschitz bound and the perturbation assertion.

For the algorithmic statement, enumerate these actions and introduce variables $\lambda_x, D_{xc}, \delta$ and $u_{gc} \geq 0$ with

$$-u_{gc} \leq b(g,c) - (\lambda P_g D)_c \leq u_{gc}, \qquad \sum_c u_{gc} \leq 2\delta.$$

Together with the stochastic constraints these are polynomial inequalities over the real algebraic numbers. The point-start branch fixes $\lambda = e_x$ and enumerates x. The compact attained minimum is the least real point of the resulting semialgebraic set of feasible δ's. Real quantifier elimination and algebraic sample-point construction, used with the same scope as Theorem S10.2, compute the value and tables. □

The finite optimization concerns synthesis against a supplied finite-group target. It does not solve the cutpoint, equivalence, or isolation problem for a supplied arbitrary probabilistic automaton. Nor does it replace the specification for all words by a check of one representative per group element on the original stochastic controller. Those restrictions would change the problem.

Theorem 4.3 (Robust multiplicity and initialization certificates)**.** *Use the stacked Fourier matrix $A_\rho(b)$ of* (S3.1)*, with singular values in decreasing order, and write $m_\rho(X)$ for the multiplicity of ρ in $\mathbb{C}[X]$. If an action X approximates b to total-variation error at most δ, then*

$$\sigma_{m_\rho(X)+1}(A_\rho(b)) \leq \sqrt{2}\,\delta \qquad (\rho \in \widehat{G}). \tag{4.6}$$

A singular value past the number of columns means zero. For a binary response $b = (\beta, 1-\beta)$ the sharper equivalent scalar certificate is

$$\sigma_{m_\rho(X)+1}(\widehat{\beta}(\rho)) \leq \delta. \tag{4.7}$$

In particular let r_ρ^δ be the number of singular values of $A_\rho(b)$ strictly exceeding $\sqrt{2}\delta$. Then every mixed-start controller of uniform error at most δ uses at least

$$\min\left\{\sum_H q_H[G:H] : q_H \in \mathbb{Z}_{\geq 0}, \quad \sum_H q_H \geq 1, \quad \sum_H q_H m_\rho(H) \geq r_\rho^\delta \ (\rho \in \widehat{G})\right\} \tag{4.8}$$

labels. The scalar version may be used for binary outputs.

Let N be the number of distinct residual tables $b^h(g) = b(hg)$. If $N > 1$, let

$$\Delta_b = \min_{b^h \neq b^k} \max_g \mathrm{TV}(b(hg), b(kg)) > 0.$$

Whenever $\delta < \Delta_b/2$, a pure-start controller with fewer than N labels must have a projected mixed preparation. Its total budget is therefore at least one plus the lower bound (4.8).

Proof. Put $a(g) = \lambda P_g D$ and $z_c(g) = b(g,c) - a(g,c)$. Since $\sum_c z_c(g) = 0$ and $\frac{1}{2}\sum_c |z_c(g)| \leq \delta$, the sums of the positive and negative parts are each at most δ, and $\sum_c |z_c(g)|^2 \leq 2\delta^2$. Unitarity gives

$$\left\| \begin{pmatrix} z_{c_1}(g)\rho(g^{-1}) \\ \vdots \\ z_{c_{|C|}}(g)\rho(g^{-1}) \end{pmatrix} \right\|_{\mathrm{op}} = \left(\sum_c |z_c(g)|^2\right)^{1/2} \leq \sqrt{2}\delta.$$

Average over g to obtain $\|A_\rho(b) - A_\rho(a)\|_{\mathrm{op}} \leq \sqrt{2}\delta$. The Fourier-multiplicity lemma gives $\operatorname{rank} A_\rho(a) \leq m_\rho(X)$. For completeness, if B has rank at most m, its nullspace intersects the span of the first $m+1$ right singular vectors of A in a nonzero vector. On a unit vector in that intersection, $\|(A-B)v\| = \|Av\| \geq \sigma_{m+1}(A)$. This proves (4.6); the binary version follows by averaging the scalar error, whose absolute value is at most δ, against unitary matrices. Summing orbit multiplicities and using the mixed-start normal form gives the integer bound.

For the last assertion, suppose the projected preparation were a point x. Its distinct residuals are indexed by points $x \cdot h$ in its orbit. If two target residuals b^h, b^k were represented by the same point, the two approximating residuals would coincide for every future g, and the triangle inequality would give distance at most 2δ between b^h and b^k. This contradicts the definition of Δ_b. The orbit must therefore contain at least N points. With fewer than N total labels this is impossible, so the projected start is mixed and its extra initializer is charged by the normal-form theorem. □

Corollary 4.4 (Uniform error cannot replace a prime orbit)**.** *Let $G = C_p$ with p prime and let $b : G \to \Delta(C)$ be any target. Put $r(b) = \min_{q \in \Delta(C)} \max_g \mathrm{TV}(b(g), q)$. For either start convention,*

$$\mathcal{E}_M(b) = \begin{cases} r(b), & 1 \leq M < p, \\ 0, & M \geq p. \end{cases}$$

For binary outputs $r(b) = (\max_g \beta(g) - \min_g \beta(g))/2$.

Proof. Every orbit of C_p has size one or p. Thus any action with fewer than p points is trivial and gives a constant response, regardless of preparation. A one-label controller attains $r(b)$. On p points the regular action with a point preparation and decoder b is exact. For binary outputs the midpoint of the smallest interval containing all values of β is a Chebyshev center of that interval. □

For the symmetric-group target (S3.6), the scalar standard block is $tI/[n(n-1)]$ and $\Delta_{b_t} = 2t/n$. Indeed a permutation moving k points changes a residual by at most kt/n, with equality at an inverse representative; the smallest nonzero k is two. The subgroup-index estimate (S3.12) and the preceding theorem imply

$$\mathcal{E}_M^{\mathrm{p}}(b_t) \geq \frac{t}{n(n-1)} \qquad (n \geq 5,\ 0 < t \leq 1,\ M \leq n(n-1)). \tag{4.9}$$

If the error were smaller, the full standard multiplicity $n-1$ would be forced, and the small-error initialization charge would give at least $n(n-1)+1$ labels; here $n! > n(n-1)$. This explicit lower

bound is not asserted to be the optimal error constant. At every exact resource jump the finite optimization also gives a strictly positive robustness radius, without a claimed closed formula for that radius.

5. Shared preparation, invariant polytopes, and noncyclic memory

A device that reproduces one prepared response need not reproduce a family of preparations with the same memory, transitions, and decoder. We now solve the structural problem with this common-device requirement. The preparation is selected before the command word; it is not revealed to the terminal decoder. In particular, one may not choose a different decoder after learning which preparation was selected. This convention is what makes joint realization a resource question.

Fix a finite group G, a surjective command homomorphism $\gamma : A^* \to G$, a finite outcome set C, and a nonempty finite preparation set I. A *shared controller* consists of one family of stochastic matrices Q_a and one stochastic decoder D_0, together with an initial row for each $i \in I$. In the pure convention these rows are prescribed label rows e_{o_i}; different preparations may use the same label when their required laws agree. All labels, including initialization labels, are counted. In the mixed convention the initial rows may be arbitrary probability rows and their external preparation is not counted. The result concerns this explicit distinction, not the bit cost of selecting i or an uncounted label made available to the decoder.

For each i, g let $K_{i,g} \subseteq \Delta(C)$ be closed and convex, possibly empty. The requirement is

$$e_{o_i} Q_w D_0 \in K_{i,\gamma(w)} \qquad (i \in I,\ w \in A^*), \tag{5.1}$$

with arbitrary initial rows in the mixed convention. For probability rows $\Lambda = (\lambda_i)_{i \in I}$ on a finite set put

$$\iota(\Lambda) = \big|\{\lambda_i : i \in I,\ \lambda_i \text{ is not a point mass}\}\big|.$$

Repeated rows are counted once.

Theorem 5.1 (Shared normal form and exact initialization charge)**.** *The specification* (5.1) *has a pure shared controller on at most M labels if and only if there are a finite G-set X, probability rows λ_i on X, and one stochastic decoder D such that*

$$\lambda_i P_g D \in K_{i,g} \quad (i \in I, g \in G), \qquad |X| + \iota(\Lambda) \le M. \tag{5.2}$$

For mixed preparations the criterion is $|X| \le M$. The action may have a kernel. Thus a single permutation core suffices for all preparations, and each distinct mixed preparation costs exactly one additional label in this normal form.

Proof. Use the identity-word construction in the proof of Theorem 4.1. Its averaging matrix T, stochastic idempotent $E = \lim_n T^n$, and disjoint closed classes $C_1, \dots, C_k$ depend only on the command matrices, not on the initial row. Write μ_x for their stationary probability rows. The sandwiched maps EQ_uE permute these rows and give one action $P_{\gamma(u)}$ on $X = \{1, \dots, k\}$. Put

$$e_{o_i} E = \sum_{x \in X} \lambda_i(x) \mu_x, \qquad D(x, \cdot) = \mu_x D_0.$$

The same limit as in that theorem gives

$$\lambda_i P_g D = \lim_{n \to \infty} e_{o_i} T^n Q_u T^n D_0 \in K_{i,g}, \qquad \gamma(u) = g,$$

because each approximant is a convex combination of laws for words representing g. There is no averaging between different preparations.

If o_i lies in a closed class, then $e_{o_i}E$ is its stationary row, so λ_i is a point mass. Distinct mixed λ_i must therefore arise from distinct initial labels outside the closed classes. These classes use at least k labels, so $\iota(\Lambda) \le M - k$. This argument also permits an outside initial label to project to a point mass, and it does not charge that label when the projected realization no longer needs it. It permits

several initial labels to have the same projected mixed row, in which case they are merged. For mixed initial rows the same projection gives $k \le M$ without the additional count.

Conversely, retain the permutation labels X. For every distinct nonpoint row λ among the λ_i, adjoin one label $\star_\lambda$ with decoder λD and, on command a, transition row $\lambda P_{\gamma(a)}$ into X. No transition returns to an adjoined label. A point row uses its own point of X as initial label; a nonpoint row uses $\star_\lambda$. The empty-word law is $\lambda_i D$ and the law after every nonempty word is $\lambda_i P_{\gamma(w)} D$. This establishes (5.2) with the exact displayed count. The mixed case starts directly from λ_i on X. □

For exact targets write $\mathcal{B} = \{b_i : i \in I\}$, where $b_i : G \to \Delta(C)$; duplicate target tables are identified. Let $\kappa_G(\mathcal{B})$ and $m_G(\mathcal{B})$ denote the mixed and pure shared minima. The first equals the least $|X|$ in a simultaneous factorization $b_i(g) = \lambda_i P_g D$; the second includes $\iota(\Lambda)$. In particular

$$\max_i \kappa_G(b_i) \le \kappa_G(\mathcal{B}), \qquad \kappa_G(\mathcal{B}) \le m_G(\mathcal{B}).$$

Neither inequality compares a collection of separately optimized decoders with a single common decoder.

5.1. **A geometric formula with all initialization labels counted.** Let $\mathcal{P}_G(\mathcal{B})$ be the set of nonempty polytopes in the finite-dimensional product simplex

$$\mathcal{Q} = \{q : G \to \Delta(C)\}$$

that contain $\mathcal{B}$ and are invariant under $(q \cdot h)(g) = q(hg)$. The whole product simplex is one such polytope. Write $v(P) = |\operatorname{vert} P|$. The polytope is allowed to extend beyond the linear span of the prescribed responses. Forbidding this extension would impose an additional positive-realization constraint.

Theorem 5.2 (Invariant-polytope resource formulas)**.** *For every nonempty finite target family,*

$$\kappa_G(\mathcal{B}) = \min_{P \in \mathcal{P}_G(\mathcal{B})} v(P), \tag{5.3}$$

$$m_G(\mathcal{B}) = \min_{P \in \mathcal{P}_G(\mathcal{B})} \left(v(P) + |\mathcal{B} \setminus \operatorname{vert} P|\right). \tag{5.4}$$

The minima are attained. Every minimizing polytope constructs an optimal controller: its vertices are permutation labels, and each distinct prescribed nonvertex response receives one transient initialization label.

Proof. Given a simultaneous factorization, associate to each core label the profile $q_x(g) = e_x P_g D$. The set of profiles is invariant under G, and $P = \operatorname{conv}\{q_x : x \in X\}$ belongs to $\mathcal{P}_G(\mathcal{B})$. Every vertex is a profile, so $v(P) \le |X|$. This proves the necessary mixed bound.

For the pure bound, apply the preceding normal form to an arbitrary M-label pure realization. Let $s = \iota(\Lambda)$ and $k = |X|$, so $k + s \le M$. If a target b_i is not a vertex of P, it either equals a nonvertex core profile q_x, or is not any core profile. There are at most $k - v(P)$ distinct profiles of the first kind: at least $v(P)$ core labels are required to represent the different vertices. Every distinct target of the second kind requires a distinct nonpoint λ_i, so there are at most s such targets. Therefore

$$v(P) + |\mathcal{B} \setminus \operatorname{vert} P| \le v(P) + (k - v(P)) + s \le M.$$

This is the step that prevents a nonextreme continuing label used by another preparation from being charged twice.

Conversely, a finite group preserving P permutes its vertices. Use these vertices as X, with $D(q, c) = q(e, c)$. Barycentric coordinates of $b_i \in P$ supply a probability row λ_i with $\lambda_i P_g D = b_i(g)$. If b_i is a vertex choose its point row. If it is not a vertex, every barycentric representation is nonpoint. Choose the same representation for identical target tables. Different target tables have

different representations, since their barycentric images differ. The normal-form construction thus costs exactly $v(P) + |\mathcal{B} \setminus \operatorname{vert} P|$ in the pure convention, and $v(P)$ in the mixed convention.

The feasible integer costs form nonempty subsets of the positive integers. Their least elements are realized by some feasible polytope; the two constructions prove the asserted equalities and attainment. □

Invariant cones in positive realization and the relation between symmetry and nonnegative factorizations are classical [48, 54, 67]. Formula (5.3) is a response-polytope form of that structure. The additional resource statement here is the exact shared-device pure-initialization term in (5.4), obtained after the common identity-word projection. It neither assumes that a supplied stochastic controller satisfies the group relations nor identifies linear rank with positive memory.

5.2. **Joint Fourier multiplicity and exact low-contrast synthesis.** Let

$$V_{\mathcal{B}} = \operatorname{span}_{\mathbb{R}}\{b_i \cdot h : i \in I, h \in G\}, \qquad d = \dim V_{\mathcal{B}}.$$

For an irreducible unitary representation ρ of degree d_ρ, use the stacked matrix $A_\rho(b_i)$ from (S3.1) and define the *joint* block

$$\mathcal{A}_\rho(\mathcal{B}) = \big[A_\rho(b_1) \ \cdots \ A_\rho(b_{|I|})\big], \qquad r_\rho(\mathcal{B}) = \operatorname{rank}_{\mathbb{C}} \mathcal{A}_\rho(\mathcal{B}). \tag{5.5}$$

Unlike the single-preparation rank, $r_\rho(\mathcal{B})$ can exceed d_ρ. The same decoder must accommodate the union of the Fourier column spaces, rather than their largest separate rank.

Lemma 5.3 (Joint multiplicity)**.** *The multiplicity of ρ in $V_{\mathcal{B}} \otimes_{\mathbb{R}} \mathbb{C}$ equals $r_\rho(\mathcal{B})$. Consequently every simultaneous factorization on a G-set X satisfies*

$$r_\rho(\mathcal{B}) \le m_\rho(X) \qquad (\rho \in \widehat{G}).$$

Proof. Fourier transformation sends the translate $b_i \cdot h$ to $A_\rho(b_i)\rho(h)$. The group-algebra decomposition used in Lemma S3.1 therefore identifies its ρ-component with

$$\left\{ \sum_{i \in I} A_\rho(b_i) M_i : M_i \in M_{d_\rho}(\mathbb{C}) \right\}.$$

The matrices M_i are independent, since the different generators may have independent group-algebra coefficients. This is the space of all matrices whose columns belong to the sum of the column spaces of the $A_\rho(b_i)$. That sum has dimension $r_\rho(\mathcal{B})$, so the right module consists of precisely that many rows of length d_ρ. For a simultaneous positive factorization, the span of all reachable preparation rows is a submodule of $\mathbb{R}^X$, and evaluation against the common decoder maps it onto $V_{\mathcal{B}}$. Complete reducibility therefore embeds $V_{\mathcal{B}}$ into $\mathbb{R}^X$, proving the bound. □

Write $\nu_G(V_{\mathcal{B}})$ for the smallest size of a permutation module containing $V_{\mathcal{B}}$. For one representative H of each subgroup conjugacy class, Frobenius reciprocity gives

$$\nu_G(V_{\mathcal{B}}) = \min \left\{ \sum_H q_H [G : H] : q_H \in \mathbb{Z}_{\ge 0}, \quad \sum_H q_H m_\rho(H) \ge r_\rho(\mathcal{B}) \ (\rho \in \widehat{G}) \right\}. \tag{5.6}$$

The equality follows by orbit decomposition and the real-descent part of Lemma S3.1; it applies also to complex and quaternionic representation types.

Theorem 5.4 (Exact noncyclic memory at small common contrast)**.** *Suppose that all preparations have the same group-averaged law,*

$$\frac{1}{|G|} \sum_{g \in G} b_i(g) = \pi \qquad (i \in I),$$

and put $b_{i,t}(g) = \pi + t\{b_i(g) - \pi\}$ for $0 \le t \le 1$. There is an explicitly positive t_0 such that

$$\kappa_G(\{b_{i,t} : i \in I\}) = \nu_G(V_{\mathcal{B}}) \qquad (0 < t \le t_0). \tag{5.7}$$

For every $t > 0$ the joint module and its Fourier multiplicities are unchanged. The exact pure resource is the invariant-polytope formula (5.4)*; in particular, if $r = |\mathcal{B}|$ and $N = |\{b_i \cdot g : i \in I, g \in G\}|$, then on this interval*

$$\nu_G(V_{\mathcal{B}}) \le m_G(\{b_{i,t}\}) \le \min\{N, \nu_G(V_{\mathcal{B}}) + r\}.$$

At $t = 0$ both resources equal one.

Proof. Remove outcomes with $\pi_c = 0$; positivity makes them identically zero for every preparation and group element. They will be restored as zero decoder columns. Let $\bar{b}$ be the constant table π. Group averaging maps every generating residual to $\bar{b}$, hence

$$V_{\mathcal{B}} = \mathbb{R}\bar{b} \oplus W, \qquad W = \ker\left(|G|^{-1} \sum_{g \in G} L_g\right).$$

The nontrivial part W has no invariant vector and all its elements have zero total output mass. If $W = 0$, all targets equal $\bar{b}$ and the assertion is immediate. Otherwise choose a minimum G-set X with $k = \nu_G(V_{\mathcal{B}})$ and a real equivariant injection $j : W \to \mathbb{R}^X$. Let $F : \mathbb{R}^X \to W$ be orthogonal projection onto $j(W)$ followed by j^{-1}. Since the permutation inner product is invariant, F is equivariant and annihilates every invariant row. Write

$$p = (1/k, \dots, 1/k), \qquad a_i = j(b_i - \bar{b}), \qquad T(x, c) = F(e_x)(e, c).$$

Then $a_i\mathbf{1} = 0$, $T\mathbf{1} = 0$, $pT = 0$, and $a_i P_g T = b_i(g) - \pi$. Put

$$A = \max_i \|a_i\|_\infty > 0, \quad \epsilon = \frac{1}{2kA}, \quad B = \max_{x,c} \frac{|T(x, c)|}{\pi_c} > 0, \quad t_0 = \min\{1, \epsilon/B\}.$$

The rows $\lambda_i = p + \epsilon a_i$ have all entries at least $1/(2k)$. For $0 \le t \le t_0$ the *common* matrix $D_t(x, c) = \pi_c + (t/\epsilon)T(x, c)$ is stochastic, and

$$\lambda_i P_g D_t = \pi + t\{b_i(g) - \pi\} = b_{i,t}(g)$$

for every preparation simultaneously. This proves the matching upper bound, with one decoder and one action.

The equivariant map that is the identity on $\mathbb{R}\bar{b}$ and multiplication by t on W is invertible for $t > 0$ and sends all residuals to their contrasted versions. Thus their joint module and all equality relations are unchanged. Lemma 5.3 and (5.6) give the lower bound. The constructed preparations have at most r distinct nonpoint rows, so their pure initialization costs at most $k + r$. Alternatively, the union of all residuals is a deterministic shared machine with N labels. At zero contrast one common constant decoder and one fixed label suffice. □

The common-mean hypothesis has mathematical content. Distinct constant preparation laws can already have positive-factorization cost larger than their linear rank; contrast does not change them. For example, with the trivial group, the four preparation rows

$$\frac{1}{2}\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

have linear rank three but require four shared mixed labels. In any nonnegative rank-one summand, at most one of the four positive diagonal entries can be covered: covering two would make both corresponding cross entries positive, whereas at least one is zero. Four summands are therefore necessary, and the displayed rows supply a four-label realization. This also explains why a separate scalar module bound for each preparation cannot replace joint positive synthesis.

5.3. **Robust joint certificates and sharp resource separations.** For weights $\alpha_i \geq 0$ with $\sum_i \alpha_i = 1$, let

$$\mathcal{A}_{\rho,\alpha}(\mathcal{B}) = [\sqrt{\alpha_1}A_\rho(b_1) \ \cdots \ \sqrt{\alpha_{|I|}}A_\rho(b_{|I|})].$$

Theorem 5.5 (Joint robust multiplicity certificate)**.** *If one mixed shared controller approximates every $b_i(\gamma(w))$ to total-variation error at most δ for every word w, there is a G-set X of size at most its memory such that, for every irreducible ρ and every such weight vector,*

$$\sigma_{m_\rho(X)+1}(\mathcal{A}_{\rho,\alpha}(\mathcal{B})) \leq \sqrt{2}\delta. \tag{5.8}$$

Hence the integer cover (5.6)*, with each r_ρ replaced by the number of singular values strictly above $\sqrt{2}\delta$ in any chosen weighted block, is a memory lower bound. Weights may be chosen separately for different irreducibles. The same lower bound applies to pure shared controllers.*

Proof. Apply Theorem 5.1 to the closed convex total-variation balls. It supplies a common permutation action and tables $a_i(g) = \lambda_i P_g D$ with the same error bound. The calculation in Theorem 4.3 gives $\|A_\rho(b_i) - A_\rho(a_i)\|_{\mathrm{op}} \leq \sqrt{2}\delta$. For block rows $z = (z_i)_i$, Cauchy–Schwarz therefore gives

$$\left\|\sum_i \sqrt{\alpha_i}\{A_\rho(b_i) - A_\rho(a_i)\}z_i\right\| \leq \sqrt{2}\delta \sum_i \sqrt{\alpha_i}\|z_i\| \leq \sqrt{2}\delta \left(\sum_i \|z_i\|^2\right)^{1/2}.$$

The weighted joint error has operator norm at most $\sqrt{2}\delta$. The approximating joint block has rank at most $m_\rho(X)$ by Lemma 5.3; zero weights can only reduce that rank. The singular-value nullspace argument used in Theorem 4.3 proves (5.8). Orbit decomposition gives the stated cover. The pure normal form has a core no larger than its total memory. □

Proposition 5.6 (Unbounded joint multiplicity for a fixed group)**.** *For every integer $q \geq 1$ there is a q-preparation experiment on $G = C_2$ with $q+1$ terminal outcomes such that each preparation alone has pure and mixed memory two, while their common-device minima are*

$$\kappa_{C_2}(\mathcal{B}) = m_{C_2}(\mathcal{B}) = 2q.$$

Its joint sign multiplicity is q, although the sign representation has degree one. All target response probabilities may be chosen rational and strictly positive. Moreover, in the explicit family below every common approximation with uniform total-variation error $\delta < \epsilon/\sqrt{2q}$ still requires at least $2q$ labels.

Proof. Take $C = \{0, 1, \ldots, q\}$, the uniform law $\pi = (q+1)^{-1}\mathbf{1}$, and $v_i = e_i - e_0$ for $1 \leq i \leq q$. For any rational $0 < \epsilon < 1/(q+1)$ put

$$b_i(e) = \pi + \epsilon v_i, \qquad b_i(s) = \pi - \epsilon v_i,$$

where s is the nonidentity element of C_2. Each individual response is nonconstant and is realized by a two-cycle with these two decoder rows, so its exact memory is two. For a shared realization the joint sign block is the $(q+1) \times q$ matrix $\epsilon[v_1 \ \cdots \ v_q]$, of rank q. Every two-cycle in a C_2-set contributes one sign copy; fixed points contribute none. At least q two-cycles, or $2q$ labels, are necessary. The disjoint union of the q displayed two-cycles attains this bound, with pure preparation i at the positive point of its own cycle.

For equal weights $\alpha_i = 1/q$, the squared nonzero singular values of the joint sign block are the eigenvalues of $(\epsilon^2/q)(I + \mathbf{1}\mathbf{1}^{\mathsf{T}})$. Its least singular value is at least $\epsilon/\sqrt{q}$ (also for $q = 1$). When $\sqrt{2}\delta < \epsilon/\sqrt{q}$, the joint robust certificate still forces q sign copies. This lower bound allows arbitrary stochastic transitions and arbitrary stochastic terminal decoders satisfying the all-word approximation requirement. □

Proposition 5.7 (An unbounded pure initialization cost)**.** *For every integer $r \geq 1$ there is an r-preparation binary experiment on C_2 with*

$$\kappa_{C_2}(\mathcal{B}) = 2, \qquad m_{C_2}(\mathcal{B}) = r + 1,$$

although every preparation separately has pure memory two.

Proof. Choose $0 < t_1 < \cdots < t_r < 1/2$ and set

$$b_i(e) = (1/2 + t_i, 1/2 - t_i), \qquad b_i(s) = (1/2 - t_i, 1/2 + t_i).$$

Use one two-cycle with decoder first coordinates $1/2 + t_r$ and $1/2 - t_r$. Appropriate mixtures of its two states realize every preparation, proving the mixed upper two; nonconstancy proves its lower. The last preparation uses the positive core point, and the other $r - 1$ preparations use distinct transient initialization labels. This gives $r + 1$ pure labels by Theorem 5.1.

For the lower bound, the r distinct empty-word laws require r distinct pure initial labels. If those were all the labels, every decoder row would have first coordinate strictly greater than $1/2$. Every stochastic transition from any of them would then still give a first-coordinate probability strictly greater than $1/2$. This contradicts the required response after s, whose first coordinate is strictly less than $1/2$. At least one further label is necessary. □

Both families are ordinary finite causal experiments, and the constructions also admit diagonal quantum realizations: place the decoder laws on the diagonal, use permutation unitaries for commands, and use the stated diagonal preparations. No quantum advantage is asserted. Their purpose is to separate simultaneous preparation, noncyclic multiplicity, and counted pure initialization from the already solved single-preparation resource. For arbitrary different mean laws, arbitrary quantum preparation sets, or nonreversible commands, the displayed common-mean equality is not asserted.

6. Primitive optimal control for finite reset instruments

An exact application should not require that a balancing group has already been built into the experiment. We give a class in which the best continuation and a least-favorable preparation can instead be derived from the primitive measurement instrument. The class includes any finite family of rank-one failure projections in any finite Hilbert dimension, as well as mixed post-failure states. It does not subsume an arbitrary-rank failure projection, whose posterior still depends on its input. The complete arbitrary-rank two-detector theorem is retained in Section S37 of the supplement.

Let $\mathcal{H}$ be finite-dimensional. There is a nonempty finite command set $\mathcal{A}$, nonterminal reports $i = 1, \ldots, s$, and a terminal success. Assume $s \geq 1$. A nonterminal branch has the physically specified map

$$\mathcal{I}_{a,i}(\rho) = \operatorname{tr}(E_{a,i}\rho)\sigma_i, \qquad E_{a,i} \succeq 0, \quad \sum_i E_{a,i} \preccurlyeq I, \tag{6.1}$$

where σ_i is a known density matrix and the report i is observed. The aggregate success effect is $B_a = I - \sum_i E_{a,i}$. The instrument itself prepares the state σ_i on failure; no additional freely available reset is assumed. One call has a nonnegative cost with conditional mean $\operatorname{tr}(G_a\rho)$, where

$$0 < c_* I \preccurlyeq G_a \preccurlyeq c^* I < \infty.$$

Success requires immediate commitment and failure forbids it. The decoder has one fixed terminal value. All commands are legal. The initial state is arbitrary and unknown to the observer; the known post-report state is part of the physical model, not an unknown-state oracle. Commands and reports may be relabelled without changing these data.

Set

$$Q_{iaj} = \operatorname{tr}(E_{a,j}\sigma_i), \qquad c_{ia} = \operatorname{tr}(G_a\sigma_i), \qquad q_{ia}^{\mathrm{suc}} = 1 - \sum_j Q_{iaj}.$$

A nonempty set $S \subset \{1, \ldots, s\}$ is a *closed failure set* when $\sum_{j \in S} Q_{iaj} = 1$ for every $i \in S$ and every command a.

Theorem 6.1 (Finite reset classification, optimum and preparation saddle)**.** *There exists one common visible-history policy which is proper at finite uniform expected cost for every initial density matrix if and only if there is no closed failure set. If there is none, the finite Bellman system*

$$v_i = \min_{a \in \mathcal{A}} \left(c_{ia} + \sum_j Q_{iaj} v_j \right), \qquad 1 \le i \le s, \tag{6.2}$$

has a unique finite nonnegative solution. It is the optimal remaining cost from each reported state σ_i. A deterministic minimizing command at every reported state attains it and is proper.

Define Hermitian start-cost operators

$$L_a = G_a + \sum_j v_j E_{a,j}.$$

The unknown-initial-state value is exactly

$$V_* = \min_{p \in \Delta(\mathcal{A})} \lambda_{\max} \left(\sum_a p_a L_a \right) \tag{6.3}$$

$$= \max_{\rho \succeq 0,\ \operatorname{tr}\rho = 1} \min_a \operatorname{tr}(\rho L_a). \tag{6.4}$$

Both extrema are attained. A minimizing initial command distribution followed by the deterministic Bellman continuation is one robustly minimax policy whose uniform bound holds for every initial state; its initial row need not minimize cost separately at each known state. A maximizing density matrix is a primitive least-favorable preparation against every visible-history policy. The latter statement is for unrestricted history; it is not automatically a saddle at every smaller memory budget.

If g distinct command values occur in a chosen deterministic Bellman selector, then $g + 1 \le \min(s, |\mathcal{A}|) + 1$ live labels suffice with a prescribed initial label. If a minimizing initial distribution is a pure command already used by that selector, g suffice. Event stopping permits reuse of a live label for the constant terminal value; state stopping adds one disjoint terminal label.

Proof. If a closed failure set exists, start in any σ_i with $i \in S$. Every command fails with probability one and reports another state in S. Induction makes success impossible under every policy; the positive cost floor makes its expected cost infinite.

Suppose there is no such set. Choose every command independently with strictly positive probability. From each failure state there is a positive-probability directed path to success: otherwise the set of failure states reachable from that state would be nonempty and closed under every command, contradicting the assumption. A shortest such path visits no failure state twice and has at most s edges. Finiteness gives a positive lower bound δ for the probability of following one such path in the next s calls, uniformly in its starting failure state. Consequently survival after ns calls is at most $(1 - \delta)^n$. From an arbitrary initial state the first call either succeeds or enters one of the reported states. The expected number of calls is therefore uniformly at most $1 + s/\delta$, and the expected cost at most $c^*(1 + s/\delta)$. This proves existence using even a one-live-label randomized command row. Its role is feasibility, not optimality.

For explicit primitive constants, if the row chooses each command with probability at least $\eta > 0$ and $\epsilon > 0$ is the smallest positive number among the finite collection $Q_{iaj}, q_{ia}^{\mathrm{suc}}$, then one may take $\delta = (\eta\epsilon)^s$. Factors for a shorter path can be bounded below by this same number. Zero probabilities are not divided out or assigned a false positive floor.

Start value iteration at $v^{(0)} = 0$. The finite-horizon dynamic programming recursion gives $v_i^{(n+1)} = \min_a(c_{ia} + \sum_j Q_{iaj} v_j^{(n)})$. These values increase and are bounded by the proper randomized policy's finite cost. Their limit v is finite and solves (6.2), since a finite minimum commutes with this coordinatewise limit. Select a minimizing command $\alpha(i)$ in each row. Iterating its equality bounds every partial expected cost by v_i. The positive floor then bounds the sum of survival probabilities,

so the selected policy is proper and has finite cost. The bounded terminal remainder $\mathbb{E}[v_{I_n}\mathbf{1}_{\{\tau>n\}}]$ vanishes. Passing to the limit in the iterated equality identifies its cost as v_i.

For any other proper policy, Bellman's inequality for each chosen command telescopes to v_i no greater than its expected cost plus the same bounded survival remainder. Letting the horizon grow proves its cost at least v_i. An infinite-cost policy cannot improve this bound. The same argument for any finite nonnegative Bellman solution w shows that its greedy policy realizes w and that w is below the cost of every proper policy. Comparing to v in both directions proves uniqueness.

Before the first call the observer has no report, so its first command has one model-independent distribution p. After a failure j, the physical state is exactly σ_j, irrespective of the original state or previous private coins. The already proved continuation lower therefore makes its initial cost at least $\operatorname{tr}[\rho\sum_a p_a L_a]$. The Bellman continuation attains this bound for every ρ. Taking the worst density matrix gives the largest eigenvalue, proving (6.3). Compactness of the probability simplex gives its minimum.

For completeness the dual equality follows directly from finite-dimensional separation. Let $\mathcal{K} = \operatorname{conv}\{L_a\}$ and let $v_* = \min_{L\in\mathcal{K}} \lambda_{\max}(L)$. The compact convex set $\mathcal{K}$ is disjoint from the open convex set $\{H : H \prec v_* I\}$. A separating real linear functional on Hermitian matrices is pairing with a nonzero Hermitian R, with $\inf_{L\in\mathcal{K}} \operatorname{tr}(RL) \geq \sup_{H\prec v_* I} \operatorname{tr}(RH)$. Finiteness of the right side forces $R \succeq 0$, since otherwise moving H arbitrarily far in a negative semidefinite direction makes that pairing unbounded above. Normalize $\operatorname{tr} R = 1$; the right supremum is then v_*. This gives a density matrix with $\min_a \operatorname{tr}(RL_a) \geq v_*$. Conversely, pairing any density matrix with a minimizing L bounds that minimum by v_*. This proves equality and dual attainment. The continuation lower makes the maximizing state a witness against every policy, not merely against first-call mixtures.

To implement the policy, use one label for its initial distribution and a pure command label for each distinct value of α. On failure report j, retain the label for $\alpha(j)$. The next row depends only on that retained label. Success commits immediately. If the initial row is already one of these pure rows, use that existing label as the start. The event/state packing lemma gives the terminal counts. No running clock, earlier report or persistent seed is available in addition to the labels. □

Proposition 6.2 (Equality, rigidity and a memory lower certificate)**.** *Let ρ_* be an attaining preparation in* (6.4)*, and put*

$$d_{ia} = c_{ia} + \sum_j Q_{iaj} v_j - v_i \geq 0, \qquad \mathcal{A}_i^* = \{a : d_{ia} = 0\}.$$

For every finite-cost policy with first-command distribution p, its cost satisfies the exact residual identity

$$\mathbb{E}_{\rho_*} C = \operatorname{tr}\left(\rho_* \sum_a p_a L_a\right) + \mathbb{E}_{\rho_*} \sum_{n\geq 1:\, \tau>n} d_{I_n, A_{n+1}}. \tag{6.5}$$

Consequently an unrestricted optimum uses only commands in $\mathcal{A}_i^*$ at every positive-probability continuation from reported state i, under this preparation.

Suppose a set I of reported states is visited with positive probability by every robust optimum evaluated at ρ_*. Let $\chi(I)$ be the least number of subsets covering I whose respective intersections of optimal command sets are nonempty. Every fixed-row autonomous optimum needs at least $\chi(I)$ live labels. If every minimizing initial row has support not contained in $\mathcal{A}_i^*$ for every $i \in I$, it needs at least $\chi(I) + 1$. These are checkable lower certificates once the mandatory set and the initial-row condition have been proved, not assertions that every experiment has the same minimal memory.

Proof. After each failed call, subtract v_{I_n} from the cost of the next call plus the conditional next potential. The conditional difference is exactly $d_{I_n, A_{n+1}}$. Sum to a deterministic horizon, including the first-call cost and potential. The remaining potential has absolute value at most $\max_i v_i$ times the survival indicator. Finite expected cost and the floor make survival tend to zero. Monotone convergence for the nonnegative residual sum yields (6.5). Its first term is at least V_* by the dual

witness. A robust optimal policy's cost at ρ_* is at most V_*, so every visited positive-probability residual vanishes.

A retained label serving states in a subset J has one command row. Its support must lie in every $\mathcal{A}_i^*$, $i \in J$, because each such visit has positive probability and any action of positive row weight with positive residual would contradict equality. Thus $\bigcap_{i\in J} \mathcal{A}_i^*$ is nonempty. The labels used by mandatory states cover I, proving the first lower. For an optimum, its first row is a minimizing row in (6.3): its worst-state cost dominates the first-call lower, whose minimum is V_*. Under the additional support condition this initial row can serve none of the mandatory states, so its label is additional to their covering labels. This proves the sharpened count. □

Corollary 6.3 (Arbitrary finite rank-one detector families)**.** *Let the failure projection of detector a be $P_a = |u_a\rangle\langle u_a|$ on any finite-dimensional Hilbert space, with $\|u_a\| = 1$, and let its deterministic call cost be $k_a > 0$. There is no intervening evolution or additional control. If all rays $\mathbb{C}u_a$ coincide, that ray is dark and the all-state value is infinite. Otherwise the value is finite and is given by Theorem 6.1, with*

$$v_i = \min_a\{k_a + |\langle u_i, u_a\rangle|^2 v_a\}, \qquad L_a = k_a I + v_a P_a.$$

Thus the continuation is a finite scalar system, while the unknown-initial-state optimization retains the full spectral geometry of the physical projections. There is no fixed prior or restriction to two detectors or two physical dimensions.

Proof. A rank-one failure satisfies $P_a\rho P_a = \mathrm{tr}(P_a\rho)P_a$, so it is a reset branch with report a and state P_a. If all rays agree, their common ray never succeeds. Otherwise choose two distinct rays and alternate their detectors. Their squared overlap is less than one. After the first failure every switch has this same strictly positive success probability, giving finite uniform cost. The classification theorem applies. Substitution of the reset transition probabilities gives the displayed formulas. The general equality and memory certificates retain the actual observed failure label and therefore apply to this family as well. □

The reduction to a finite absorbing decision problem, Bellman verification, convex separation and spectral duality are classical methods; quantum decision-process frameworks already provide much broader control languages [12, 13, 14]. The contribution claimed here is the explicit primitive reset-instrument criterion, the common all-initial-state spectral value, its realization with charged rows, and the residual-based memory certificate within this experiment. It is distinct from optimizing the probing interval for a fixed detector with unitary evolution [19, 20]. It also does not replace the preserved two-stage reset communication and rank-constrained discrimination results by a first-detection theorem.

7. Exact optimal memory for reset experiments with tied actions

The realization of a prescribed response and the implementation of an optimal feedback rule are different problems. We now give an exact memory classification for a class of controlled experiments in which optimal actions may be tied. The class allows any finite command set, any finite number of reset reports, arbitrary finite Hilbert dimension, state-dependent call costs, and an unknown initial density matrix. The additional hypothesis is strict positivity of each failure effect. It supplies the reachability needed for a sharp converse; it is not a uniqueness assumption on an optimal decision.

Use the reset instrument of Section 6:

$$\mathcal{I}_{a,i}(\rho) = \mathrm{tr}(E_{a,i}\rho)\sigma_i, \qquad \sum_{i=1}^{s} E_{a,i} \preccurlyeq I, \qquad c_* I \preccurlyeq G_a \preccurlyeq c^* I.$$

Here $a \in \mathcal{A}$, $1 \le i \le s$, $c_* > 0$, and success requires immediate commitment. There is no additional state measurement or free reset. Throughout this section assume

$$E_{a,i} \succ 0 \quad (a \in \mathcal{A},\ 1 \le i \le s), \qquad \text{there is no closed failure set.} \tag{7.1}$$

Theorem 6.1 supplies the finite Bellman vector v, the start operators $L_a = G_a + \sum_i v_i E_{a,i}$, and the value

$$V_* = \min_{p\in\Delta(\mathcal{A})} \lambda_{\max}\Big(\sum_a p_a L_a\Big).$$

Define the Bellman residuals and optimal action sets by

$$d_{ia} = \operatorname{tr}(G_a\sigma_i) + \sum_j \operatorname{tr}(E_{a,j}\sigma_i)v_j - v_i, \qquad \mathcal{A}_i^* = \{a : d_{ia} = 0\}.$$

Every d_{ia} is nonnegative and every $\mathcal{A}_i^*$ is nonempty. Put

$$\mathcal{P}_* = \left\{p \in \Delta(\mathcal{A}) : \lambda_{\max}\Big(\sum_a p_a L_a\Big) = V_*\right\}, \qquad I(p) = \{i : \operatorname{supp} p \subseteq \mathcal{A}_i^*\}.$$

For $J \subseteq \{1,\dots,s\}$ let

$$t(J) = \min\{|H| : H \subseteq \mathcal{A},\ H \cap \mathcal{A}_i^* \neq \varnothing\ (i \in J)\}, \qquad t(\varnothing) = 0. \tag{7.2}$$

This is an ordinary transversal number of the displayed finite set system. It counts distinct continuing command rows, rather than physical states or the support size of a least-favorable preparation.

Theorem 7.1 (Exact optimal-memory formula with ties)**.** *Under* (7.1)*, the least number of live labels of a fixed-program, prescribed-pure-start autonomous controller attaining the all-initial-state optimum V_* is*

$$m_* = 1 + \min_{p\in\mathcal{P}_*} t\big(\{1,\dots,s\} \setminus I(p)\big). \tag{7.3}$$

Every minimum is realized by one optimal initial command row and pure continuing command rows. The post-report transition may be chosen as a deterministic function of the current reset report alone. Arbitrary stochastic transitions, dependence on the previous label and command, and mixed continuing rows cannot lower (7.3)*.*

Event stopping has the same total-label minimum. State stopping has total-label minimum $m_* + 1$. For algebraic primitive data the value, optimal label count, and one realizing controller can be determined by a finite exact procedure; polynomial running time is not asserted.

Proof. Choose an attaining dual density matrix ρ_* from Theorem 6.1. Consider any robust optimal controller, write its fixed command rows as $p^1,\dots,p^M$, and index its pure initial label by 1. Its cost from every ρ is at least $\operatorname{tr}(\rho\sum_a p_a^1 L_a)$. Consequently $\lambda_{\max}(\sum_a p_a^1 L_a) \le V_*$, and the reverse inequality holds by the definition of V_*. Thus $p^1 \in \mathcal{P}_*$.

The Bellman residual identity of Proposition 6.2 holds at ρ_*. Its initial term is at least V_*, whereas the whole cost is at most V_*. Every nonnegative residual on a reached continuation therefore vanishes. For every reset report i, its probability on the first call is

$$\sum_a p_a^1 \operatorname{tr}(E_{a,i}\rho_*) > 0.$$

This strict inequality follows from positive definiteness and holds even if ρ_* is singular or p^1 has proper support. After this report at least one label k is reached with positive probability. The next command has row p^k, so $\sum_a p_a^k d_{ia} = 0$. Nonnegativity implies $\operatorname{supp} p^k \subseteq \mathcal{A}_i^*$.

If $i \notin I(p^1)$, that label cannot be 1. For each other label which serves one or more such states, choose any action in the support of its row. It belongs to every optimal action set served by that row. These at most $M-1$ chosen actions meet all $\mathcal{A}_i^*$ with $i \notin I(p^1)$. This proves the lower bound in (7.3). It uses only first-failure events; no reachability conclusion is inferred from an unrelated path law.

Conversely, fix $p \in \mathcal{P}_*$ and a smallest transversal H of the remaining states. Use one initial label with command row p and one pure command label for each $a \in H$. After a failure report i, return to the initial label when $i \in I(p)$; otherwise select a fixed $a \in H \cap \mathcal{A}_i^*$. Every continuing row is then Bellman optimal for its actual reset state. Bellman's equality bounds every finite-prefix cost

from σ_i by v_i. The positive cost floor implies finite expected duration, and the bounded remaining potential vanishes. Hence the continuation cost is exactly v_i. From an arbitrary initial state the full cost is $\operatorname{tr}(\rho \sum_a p_a L_a) \le V_*$. This is a proper robust optimum. There are finitely many supports of p, so the integer minimum in (7.3) is attained. Reusing a live label for the single terminal decoder is permitted exactly in the event-stopped interface, giving the claimed terminal counts.

For the effective assertion enumerate deterministic command selectors on the s reset states. A proper selector has cost $(I - Q_\alpha)^{-1} c_\alpha$; properness and this vector can be checked by finite algebraic operations and spectral or graph tests. Some proper selector is simultaneously optimal by Theorem 6.1, so the enumeration determines v. The spectral minimization defining V_* is a compact semialgebraic optimization. For each nonempty support $T \subseteq \mathcal{A}$, the assertion that some $p \in \mathcal{P}_*$ has exactly that support is a first-order real-algebraic condition. For example, impose $p_a > 0$ on T, $p_a = 0$ off T, $\sum_a p_a = 1$, and $\sum_a p_a L_a \preccurlyeq V_* I$; positivity of a Hermitian matrix is expressed by its principal minors after separating real and imaginary parts. Exact elimination decides this condition and supplies a sample when it is true. Finite transversal enumeration then gives the minimum and the stated implementation. The procedure does not assume that a numerical optimizer has certified a tie. □

The positive-effect hypothesis can be weakened to the existence of one attaining ρ_* for which $\operatorname{tr}(E_{a,i}\rho_*) > 0$ for all a, i; the same proof applies. The stated version is a direct primitive condition, without an unknown optimal-trajectory assumption. When some reports can be inaccessible, the formula for all s reset states need not be necessary: an unreachable state may carry an arbitrary optimal action set. In that case the lower certificates of Proposition 6.2 retain their original scope.

7.1. **A finite gap certificate for every insufficient memory.** For $M \ge 1$ and rows $p^1, \dots, p^M \in \Delta(\mathcal{A})$ put

$$r_i(p^1, \dots, p^M) = \min_{1 \le k \le M} \sum_a p_a^k d_{ia}, \qquad K_M(p^1, \dots, p^M) = \sum_a p_a^1 L_a + \sum_{a,i} p_a^1 r_i(p^1, \dots, p^M) E_{a,i},$$

and define the finite variational quantity

$$D_M = \min_{(p^1, \dots, p^M) \in \Delta(\mathcal{A})^M} \lambda_{\max}(K_M(p^1, \dots, p^M)) - V_*. \tag{7.4}$$

Theorem 7.2 (Attained certificate of a strict optimality gap)**.** *Under* (7.1)*, every robustly proper controller with at most M live labels has worst-initial-state expected cost at least $V_* + D_M$. The minimum defining D_M is attained, and*

$$D_M > 0 \quad \Longleftrightarrow \quad M < m_*.$$

For algebraic primitive data, D_M is computable by exact real algebraic optimization. It is a lower certificate, not a claimed formula for the entire memory-constrained cost frontier.

Proof. For the fixed rows of a controller, condition on the first command and its failure report i. Whichever label the controller selects, the expected Bellman residual on its next call is at least r_i. All later residuals are nonnegative. The residual identity, now at any initial ρ, gives cost at least $\operatorname{tr}(\rho K_M)$, and maximization over density matrices gives $\lambda_{\max}(K_M)$. A controller of infinite worst-state cost satisfies the lower bound trivially.

The function in (7.4) is continuous on a compact product of simplexes. Moreover $K_M \succeq \sum_a p_a^1 L_a$, so $D_M \ge 0$. If $D_M = 0$, take minimizing rows. Then $p^1 \in \mathcal{P}_*$. Evaluate K_M at an attaining dual ρ_*. Its first term is at least V_* and each coefficient $\sum_a p_a^1 \operatorname{tr}(E_{a,i}\rho_*)$ is strictly positive. Since its value is at most $\lambda_{\max}(K_M) = V_*$, every r_i is zero. Thus every reset state has at least one row supported on its optimal action set. The construction in Theorem 7.1 implements an optimal controller with these rows, hence $M \ge m_*$. Conversely an optimal controller supplied there has $r_i = 0$ for all i, so $D_M = 0$ whenever $M \ge m_*$. Compactness makes the remaining cases strictly

positive. Finite minima, products and Hermitian eigenvalue inequalities are semialgebraic, proving the last assertion. □

7.2. **Every finite optimal-action cover has a physical realization.** The preceding set system is not merely an artifact of a bound. Arbitrary finite covering problems occur in reset instruments with commuting cost and effect operators with strictly positive failure effects and constant success probability.

Theorem 7.3 (Universality of the optimal-memory covering problem)**.** *Let $q \geq 2$ and let $A_1, \dots, A_s$ be nonempty proper subsets of $\{1, \dots, q\}$. Let t be their transversal number. For every fixed rational $0 < r < 1$ there is an explicit rational reset instrument with commuting cost and effect operators in dimension $s + q$, with q commands, s failure reports, success probability $1 - r$ at every call, and costs in $[1, 4]$, such that*

$$V_* = 3 + \frac{r}{1-r} + \frac{1}{q}, \qquad m_* = t + 1.$$

Its unique optimal initial command row is uniform. The minimum is over all stochastic autonomous controllers with a prescribed pure initial label, and includes arbitrary randomized post-report updates. Thus determining optimal memory is NP-hard even on this class, while the unrestricted value and an optimal command distribution are explicit.

Proof. Take an orthonormal basis $e_1, \dots, e_s, f_1, \dots, f_q$ and set

$$\sigma_i = |e_i\rangle\langle e_i|, \qquad E_{a,i} = \frac{r}{s} I.$$

The failure map is $\mathcal{I}_{a,i}(\rho) = (r/s)\,\mathrm{tr}(\rho)\sigma_i$. For instance its Kraus operators are $\sqrt{r/s}\,|e_i\rangle\langle b|$ for all displayed basis vectors b. Add any success measure-and-prepare branch with effect $(1 - r)I$. Their sum is trace preserving, so these are actual quantum instruments. They commute at the level of the relevant effects and all post-failure states are classical basis states.

Specify the positive diagonal cost operator G_a by

$$\langle e_i, G_a e_i\rangle = 1 + \mathbf{1}_{\{a \notin A_i\}}, \qquad \langle f_j, G_a f_j\rangle = 3 + \mathbf{1}_{\{a=j\}}.$$

These means are physically realizable without extra controller information. In each call measure the displayed diagonal basis, charge its indicated cost, then use fresh coins to choose success or one of the reset reports and prepare the corresponding state. The intermediate basis outcome and realized cost are not an additional observation at the control port. Averaging them gives exactly the declared instrument and mean cost; no coherence-dependent hidden assumption is needed.

All commands have $Q_{iaj} = r/s$. Since every A_i is nonempty, $v_i = (1 - r)^{-1}$ solves the Bellman equations and is their finite solution. Its optimal action set at i is exactly A_i. The start operator is $L_a = G_a + r(1 - r)^{-1} I$. For any initial command row p, its eigenvalues on the e_i coordinates are

$$\frac{1}{1-r} + p(A_i^c) \leq 2 + \frac{r}{1-r},$$

whereas its eigenvalues on the f_j coordinates are $3 + r(1 - r)^{-1} + p_j$. The latter strictly dominate, hence

$$\lambda_{\max}\Big(\sum_a p_a L_a\Big) = 3 + \frac{r}{1-r} + \max_j p_j.$$

Its unique minimum is the uniform row, with the displayed value. Its full support lies in none of the proper sets A_i, so Theorem 7.1 gives $m_* = 1 + t$ and constructs an attaining controller.

The matrices have dimension and rational encoding length polynomial in the incidence table of the set system; one may fix $r = 1/2$. Minimum transversal is the classical covering problem [84]. Restriction to proper nonempty subsets does not remove its hardness: adjoining an unused action makes every nonempty set proper without changing its transversal number. An oracle for the optimal label number in the displayed physical family would therefore return the transversal

number by subtracting one. This proves the complexity assertion by an explicit reduction, not by a general complexity claim about POMDPs. □

Example 7.4 (Why positive reachability cannot be dropped)**.** On $\mathbb{C}^2$ take one reset state $\sigma = |0\rangle\langle 0|$, two commands a, b, and

$$E_a = E_b = \tfrac{1}{2}|0\rangle\langle 0|, \qquad G_a = \operatorname{diag}(1, 6), \qquad G_b = \operatorname{diag}(2, 5).$$

There is no closed failure set. The Bellman value is $v = 2$ and its unique optimal continuation command is a. The start operators are $L_a = \operatorname{diag}(2, 6)$ and $L_b = \operatorname{diag}(3, 5)$, so $V_* = 5$ and the unique minimizing initial row is the pure command b. Applying (7.3) without positive reachability would give two labels. Nevertheless the constant b controller has cost operator $\operatorname{diag}(4, 5)$: from σ it pays 2 per call with failure probability $1/2$, and from $|1\rangle$ it pays 5 and succeeds immediately. It is a one-label robust optimum. The least-favorable preparation $|1\rangle\langle 1|$ never sees the reset report. Thus a locally nonoptimal continuation can be hidden below the robust optimum when the exposing preparation cannot reach it. This verifies the need for the reachability hypothesis by exact primitive data.

For example, the three optimal sets $\{1, 2\}$, $\{2, 3\}$ and $\{1, 3\}$ give $m_* = 3$, although every report state has two optimal commands. Pairwise uniqueness of Bayes actions plays no role. More generally the count can grow with an arbitrary finite covering instance while the success probability remains fixed and every failure effect remains strictly positive.

Bellman verification and shortest-path optimization are classical [81]; search over finite stochastic controllers also has an extensive history [82, 83]. The assertions proved here concern the exact counted-row invariant, its initializer interaction, the strict finite gap certificate and its realization inside the stated reset experiment. They do not assert a general classification of partially observed control or replace the distinct squared-sign minimax-risk problem.

8. Finite memory certificates without physical decoder assumptions

The physical-dictionary lower bound uses the geometry of density matrices at individual hidden labels. An arbitrary classical simulator need not have that geometry. We now give a different certificate, valid for arbitrary classical decoder tables. Its conclusion is quantitatively weaker, but applies to the original algebraic qubit instrument without an unproved Diophantine hypothesis. More generally it applies whenever experimentally accessible preparations have separated exposing tests.

Nonnegative factorizations, support antichains and their relation to memory and extension complexity are classical. In particular the exact zero-diagonal support bound is already in the literature on nonnegative rank; see [67, Section 4.1 and Corollary 4.13] and [85, Section 3.1]. The argument below supplies an explicit error margin and derives its hypotheses from joint stopped records. It does not treat an approximate zero as an exact zero, or assume that a rare conditional distribution inherits the unconditional approximation error.

8.1. **A robust support certificate.**

For $J \geq 2$ define the antichain threshold

$$a(J) = \min\left\{ m \geq 1 : \binom{m}{\lfloor m/2 \rfloor} \geq J \right\}. \tag{8.1}$$

In particular $\lceil \log_2 J \rceil \leq a(J) \leq J$.

Lemma 8.1 (Robust antichain factorization certificate)**.** *Let $\lambda_1, \dots, \lambda_J$ be probability rows on M labels, and let $d_1, \dots, d_J \in [0, 1]^M$ be arbitrary columns. If*

$$\lambda_n d_n \leq \epsilon, \qquad \lambda_n d_m \geq b \quad (n \neq m), \qquad 0 < b \leq 1, \qquad \epsilon < \frac{b^2}{2M^2}, \tag{8.2}$$

then $J \le \binom{M}{\lfloor M/2 \rfloor}$. *The conclusion uses no linearity, noncontextuality or quantum representation of the decoder columns.*

Proof. Put $S_n = \{i : \lambda_{n,i} \ge b/(2M)\}$. For $n \ne m$, the nonnegative sum $\lambda_n d_m \ge b$ has a summand $\lambda_{n,i} d_m(i) \ge b/M$. Since both factors are at most one,

$$\lambda_{n,i} \ge b/M, \qquad d_m(i) \ge b/M.$$

Thus $i \in S_n$. On the other hand

$$\lambda_{m,i} \le \epsilon/d_m(i) \le M\epsilon/b < b/(2M),$$

so $i \notin S_m$. Every ordered pair of distinct sets has this witness, and the S_n form an antichain in the subsets of $[M]$.

For completeness, choose a uniformly random permutation of $[M]$. For a fixed set S of size k, the probability that its elements are the first k elements of this permutation, in any order, is $1/\binom{M}{k}$. The corresponding events for members of an antichain are disjoint, since the initial sets of one permutation form a chain. Hence

$$\frac{J}{\binom{M}{\lfloor M/2 \rfloor}} \le \sum_{n=1}^{J} \binom{M}{|S_n|}^{-1} \le 1,$$

as required. □

The combinatorial conclusion is sharp at the level of these hypotheses. For $M \ge 2$, let $k = \lfloor M/2 \rfloor$, index preparations and tests by all k-element subsets S of $[M]$, and put

$$\lambda_S(i) = \frac{\mathbf{1}_{\{i \in S\}}}{k}, \qquad d_T(i) = \mathbf{1}_{\{i \notin T\}}.$$

Then $\lambda_S d_S = 0$ and $\lambda_S d_T \ge 1/k$ for $S \ne T$, with exactly $\binom{M}{k}$ preparations. This construction concerns the abstract preparation–test table. It does not construct a small controller for the qubit's chronological command–report dynamics.

8.2. **An operational certificate from joint records.** Consider any causal experiment with J preparation procedures. The nth procedure either declares failure or ends at an observable herald H_n. Only after that herald is a binary terminal test $m \in [J]$ requested. The choice of m, including a policy's future test code, is not an input to the simulator before the herald. Its terminal decoder is one fixed table for each requested test, independent of the earlier preparation procedure except through the simulator's current counted label. These are the usual causal finite-memory conventions; allowing a decoder to inspect an uncounted copy of the entire past would define a different resource.

Let p_n be the target probability of H_n, and let f_{nm} be the target rejection probability conditional on H_n and test m. Suppose that every one of these preparation–test experiments has target expected cost at most B, and that

$$p_n \ge p_* > 0, \qquad f_{nn} = 0, \qquad f_{nm} \ge s > 0 \quad (n \ne m). \tag{8.3}$$

The terminal tests and preparation procedures may be classical or quantum; only their observable probabilities enter the next theorem.

Theorem 8.2 (Memory lower bound from approximately exposing tests)**.** *Suppose* (8.3) *holds, with* $J \ge 2$ *and* $0 < s \le 1$. *An* M*-label classical simulator that approximates the complete stopped record of every specified experiment to total-variation error at most* δ *satisfies*

$$\delta \le \frac{p_* s^2}{32M^2} \quad \Longrightarrow \quad J \le \binom{M}{\lfloor M/2 \rfloor}. \tag{8.4}$$

Consequently

$$\delta \le \frac{p_* s^2}{32J^2} \quad \Longrightarrow \quad M \ge a(J). \tag{8.5}$$

Both assertions allow freely mixed or counted pure starts and arbitrary stochastic terminal decoder tables. They require only the finite collection of J^2 displayed preparation–test experiments.

Proof. For the nth procedure let q_n be the simulator's herald probability, and let λ_n be its conditional distribution on its counted labels at that herald. It does not depend on the later requested m. Write $d_m(i) \in [0,1]$ for the rejection probability of test m at label i. Total variation controls both the herald event and the joint herald-and-rejection event, so

$$|q_n - p_n| \le \delta, \qquad |q_n \lambda_n d_m - p_n f_{nm}| \le \delta. \tag{8.6}$$

Under the hypothesis in (8.4), $\delta/p_n \le s^2/(32M^2) \le 1/32$, so $q_n > 0$ and

$$\lambda_n d_n \le \frac{\delta}{p_n - \delta} \le \frac{2\delta}{p_*} \le \frac{s^2}{16M^2}.$$

For $m \ne n$ put $t = \delta/p_n \le s^2/(32M^2) \le s/4$. Equation (8.6) gives

$$\lambda_n d_m \ge \frac{p_n s - \delta}{p_n + \delta} = \frac{s - t}{1 + t} \ge \frac{s}{2}.$$

Apply Lemma 8.1 with $b = s/2$ and $\epsilon = s^2/(16M^2) < b^2/(2M^2)$. This proves (8.4). If $M \ge J$, then $M \ge a(J)$ already. Otherwise the hypothesis in (8.5) implies the one in (8.4), and gives the same conclusion. No conditional error estimate was presumed before (8.6); normalization was performed only after the herald mass was controlled. □

This theorem keeps the cost budget in the target law. It makes no assertion about the simulator's expected duration. The hidden label can have any classical interpretation, and the decoder for one binary POVM can be unrelated to decoders for other POVMs.

8.3. **An unrestricted lower bound for the original algebraic instrument.** For the undamped qubit instrument (S6.1), fix one commanded Bloch angle θ and put $z = e^{i\theta}$ and $\omega = \theta/(2\pi)$. Let $N_{\mathrm{cl}}^{\mathrm{m}}(\delta; B)$ and $N_{\mathrm{cl}}^{\mathrm{p}}(\delta; B)$ be the least label counts approximating every common visible stopping and terminal-POVM policy of target expected calls at most B, with arbitrary classical terminal decoder tables. Transitions are fixed independently of later terminal requests. Thus these classes include the physical dictionaries of Section S8.4, but impose no Born representation on their individual labels.

Fix $B > 0$ and write $\zeta = \min\{1, 7B/8\}$. For $J \ge 2$ and a phase which is not a root of unity put

$$s_J = \min_{1 \le q < J} \frac{1 - \mathrm{Re}(z^q)}{2} > 0. \tag{8.7}$$

Theorem 8.3 (Unrestricted finite certificates and arithmetic rates)**.** *For either start convention, the following statements hold.*

(i) *For every non-root phase and every integer $J \ge 2$,*

$$0 < \delta \le \frac{\zeta\, 8^{-J} s_J^2}{32J^2} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \ge a(J). \tag{8.8}$$

This lower bound uses J terminal projectors and finitely many preparation prefixes, with no assumption on individual decoder rows.

(ii) *At the original algebraic phase $z = (3 + 4i)/5$, the explicit sufficient condition is*

$$0 < \delta \le \frac{25\zeta}{128 \cdot 200^J J^2} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \ge a(J). \tag{8.9}$$

No badly-approximable assertion about $\theta/(2\pi)$ is required.

(iii) *If $\theta = 2\pi\omega$ and $\|q\omega\|_{\mathbb{R}/\mathbb{Z}} \geq c/q$ for every $q \geq 1$, then another explicit sufficient condition is*

$$0 < \delta \leq \frac{\zeta c^4}{2 \cdot 8^J J^6} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \geq a(J). \tag{8.10}$$

In case (ii)*, and also in case* (iii)*,*

$$\liminf_{\delta \downarrow 0} \frac{N_{\mathrm{cl}}^{\mathrm{m}}(\delta; B)}{\log_2 \log_8(1/\delta)} \geq 1, \qquad \liminf_{\delta \downarrow 0} \frac{N_{\mathrm{cl}}^{\mathrm{p}}(\delta; B)}{\log_2 \log_8(1/\delta)} \geq 1. \tag{8.11}$$

The inherited physical constructions give, at the same fixed phase, $N_{\mathrm{cl}}^{\mathrm{m}} \leq K_\omega + L - 1$ and $N_{\mathrm{cl}}^{\mathrm{p}} \leq K_\omega + L$, where L is (S8.9)*; at $z = (3+4i)/5$ one may take $K_\omega = 7$. The unrestricted lower and upper therefore leave a gap between logarithmic-in-logarithm and logarithmic growth. A matching logarithmic lower is asserted only for the physical-dictionary class under the hypotheses of Theorem S8.4.*

Proof. For $n = 0, \ldots, J-1$, begin with probability ζ and otherwise stop with no call. If begun, follow the report prefix $1\,0^n$, stopping immediately on a different report. Success is the herald H_n. The target probability and conditional terminal state are

$$p_n = \zeta 8^{-(n+1)}, \qquad \rho_n = U_\theta^n \tau (U_\theta^*)^n.$$

In the real Bloch plane ρ_n has vector $(\cos(n\theta), \sin(n\theta))$. The target expected number of calls, before the cost-free terminal test, is

$$\zeta \sum_{j=0}^{n} 8^{-j} \leq \zeta\, 8/7 \leq B.$$

After success request the binary terminal POVM $(\rho_m, I - \rho_m)$, and call its second outcome rejection. Its target conditional rejection probability is

$$f_{nm} = 1 - \mathrm{tr}(\rho_n \rho_m) = \frac{1 - \cos((n-m)\theta)}{2}.$$

Thus $f_{nn} = 0$, $f_{nm} \geq s_J$ for $n \neq m$, and $p_n \geq p_* := \zeta 8^{-J}$. These are the hypotheses of Theorem 8.2, which gives (i). The simulator's terminal projector request is made only after the prefix. Causality is precisely what gives one common label distribution λ_n for all its J possible terminal tests.

For (ii), write

$$z^q = \frac{A_q + iB_q}{5^q}, \qquad A_q, B_q \in \mathbb{Z}.$$

The phase is not a root of unity: otherwise $z + z^{-1} = 6/5$ would be both a rational number and an algebraic integer, hence an integer. Consequently $\mathrm{Re}\, z^q < 1$ for every $q \geq 1$, and $5^q - A_q$ is a positive integer. Therefore

$$\frac{1 - \mathrm{Re}\, z^q}{2} = \frac{5^q - A_q}{2 \cdot 5^q} \geq \frac{1}{2 \cdot 5^q}, \qquad s_J \geq \frac{1}{2 \cdot 5^{J-1}}.$$

Substitution in (i) gives

$$\frac{\zeta 8^{-J}}{32 J^2} \frac{1}{4 \cdot 25^{J-1}} = \frac{25\zeta}{128 \cdot 200^J J^2},$$

which proves the claimed rational certificate. The phase here is the Bloch rotation, as in (S6.1); its unitary uses the half-angle. No angle is squared in this calculation.

For (iii), $\sin(\pi t) \geq 2t$ for $0 \leq t \leq 1/2$ gives, for $1 \leq q < J$,

$$\frac{1 - \cos(2\pi q\omega)}{2} = \sin^2(\pi q\omega) \geq 4\|q\omega\|_{\mathbb{R}/\mathbb{Z}}^2 \geq \frac{4c^2}{q^2} \geq \frac{4c^2}{J^2}.$$

Part (i) then yields (8.10).

For fixed B, the sufficient condition in (ii) holds with $J = \log_{200}(1/\delta) - O_B(\log\log(1/\delta))$ chosen as an integer slightly below this expression. The corresponding choice in (iii) is $J = \log_8(1/\delta) - O_{B,c}(\log\log(1/\delta))$. Since $a(J) \geq \log_2 J$, both imply (8.11). The asserted uppers are the unchanged

orbit-prefix constructions of Theorem S8.4 and its final algebraic-phase paragraph, which are also admissible in the unrestricted class. □

In fact the exact integer lower $a(J)$ retains a second-order term: the central-binomial estimate $\binom{m}{\lfloor m/2 \rfloor} = \Theta(2^m/\sqrt{m})$ gives $a(J) = \log_2 J + \frac{1}{2}\log_2 \log_2 J + O(1)$. This is a refinement of the support certificate, not an assertion of the optimal asymptotic simulator size. The lower uses terminal projector requests, whereas the earlier exact infinite-memory obstruction already used report words alone. These are different sets of operational probes. No positive extra initializer charge is deduced for the unrestricted approximation problem.

Corollary 8.4 (Every non-root algebraic phase has the same lower order)**.** *Let z be algebraic with $|z| = 1$ and not a root of unity. Let its degree be D, choose an integer $A \geq 1$ such that Az is an algebraic integer, and enumerate its conjugates as $z = z_1, z_2, \ldots, z_D$. Choose any supplied real bound*

$$H \geq \max\left\{2, A^D \prod_{j=2}^{D} \max(1, |z_j|)\right\}.$$

At that fixed phase, for either start convention,

$$0 < \delta \leq \frac{\zeta H^4}{32 \cdot 16^D \, (8H^4)^J J^2} \quad \Longrightarrow \quad N_{\mathrm{cl}}(\delta; B) \geq a(J). \tag{8.12}$$

In particular (8.11) *holds. No badly-approximable premise is necessary for this lower order.*

Proof. The number $A^q(z^q - 1) = (Az)^q - A^q$ is a nonzero algebraic integer. Its norm is a nonzero integer, so

$$1 \leq A^{qD}|z^q - 1| \prod_{j=2}^{D} |z_j^q - 1| \leq 2^{D-1} H^q |z^q - 1|.$$

Because $|z| = 1$,

$$\frac{1 - \operatorname{Re} z^q}{2} = \frac{|1 - z^q|^2}{4} \geq 4^{-D} H^{-2q}.$$

It follows that $s_J \geq 4^{-D} H^{-2(J-1)}$. Substitute this in (8.8) to obtain (8.12). A choice $J = \log_{8H^4}(1/\delta) - O_{B,H,D}(\log\log(1/\delta))$ then gives the claimed lower order. This elementary algebraic norm bound does not supply the stronger $s_J \asymp J^{-2}$ bound used for the sharp physical-dictionary logarithmic law. □

9. The exact cost of initializing a stationary response

The distinction between a freely chosen initial distribution and a prescribed initial label is not peculiar to a periodic example. There is an exact general law. Its proof is an elementary elimination in the space of future response functions, within the classical positive realization framework [48, 47, 76]. It does not require hidden states to lie in the minimal linear realization, or require their unobserved transition matrix to be irreducible.

Let A be a nonempty finite command alphabet, with a finite report set Y_a for each $a \in A$. A finite classical realization consists of nonnegative matrices $T_{a,y}$ such that

$$T_a := \sum_{y \in Y_a} T_{a,y}, \qquad T_a \mathbf{1} = \mathbf{1},$$

and an initial probability row μ. For a command–report word $w = (a_1, y_1) \cdots (a_n, y_n)$ write $T_w = T_{a_1,y_1} \cdots T_{a_n,y_n}$ and $p(w) = \mu T_w \mathbf{1}$. Optional terminal readouts have decoder columns $d_{b,z} \geq 0$, with $\sum_z d_{b,z} = \mathbf{1}$, and response $p(w; b, z) = \mu T_w d_{b,z}$. All following equalities include every specified terminal readout when these are part of the experiment.

Call the prepared response *predictively stationary under* ν if ν is a probability row on A with every $\nu(a) > 0$ and

$$\sum_{a\in A} \nu(a) \sum_{y\in Y_a} p((a,y)w) = p(w) \qquad \text{for every future word } w. \tag{9.1}$$

When terminal readouts are specified, additionally require

$$\sum_{a\in A} \nu(a) \sum_{y\in Y_a} p((a,y)w; b, z) = p(w; b, z) \qquad \text{for all } w, b, z, \tag{9.2}$$

including the empty word w. This is an equality of observable responses; it does not assume $\mu \sum_a \nu(a) T_a = \mu$ in a particular hidden realization. For one command and no terminal readout it is ordinary stationarity. A response has a one-label realization precisely when its report probabilities factor as $p(w) = \prod_j p_{a_j}(y_j)$, and every terminal readout has a fixed distribution independent of the preceding word. For one command this is the i.i.d. case.

If a stationary hidden row is desired, it can be chosen without adding labels. Put $\overline{T} = \sum_a \nu(a) T_a$. Every $\mu \overline{T}^j$ has the same future response by the displayed stationarity identities. A subsequential limit of the Cesàro averages $H^{-1} \sum_{j=0}^{H-1} \mu \overline{T}^j$ is stationary for $\overline{T}$ and retains that response. Thus the freely mixed minimum below agrees with the convention requiring a stationary hidden row for the averaged command, and with the usual stationary HMM convention when there is only one command.

Let $n_{\mathrm{mix}}(p)$ be the minimum number of labels with an arbitrary initial probability row, and let $n_{\mathrm{pure}}(p)$ be the minimum when one prescribed initial label is counted. The minima are $+\infty$ when there is no finite realization.

Theorem 9.1 (Stationary initialization law)**.** *Suppose* (9.1) *holds, together with* (9.2) *when terminal readouts are included. If the response has a one-label realization, both minima are one. Otherwise*

$$n_{\mathrm{pure}}(p) = n_{\mathrm{mix}}(p) + 1, \qquad (+\infty) + 1 := +\infty. \tag{9.3}$$

The lower holds for all real nonnegative transition tables, including reducible and nonminimal linear realizations. The upper processes the first actual command while entering the core; it neither discards a physical call nor inserts an uncounted preliminary randomization.

Proof. Take any finite pure-start realization, with initial label o, and let f_i denote the complete future response from hidden label i. Thus $f_i(w) = e_i^{\mathsf{T}} T_w \mathbf{1}$, with the analogous values for terminal readouts. Put $\overline{T} = \sum_a \nu(a) T_a$. Stationarity gives the functional identity

$$f_o = \sum_j \overline{T}_{oj} f_j. \tag{9.4}$$

If $\overline{T}_{oo} = 1$, nonnegativity and full support of ν imply $(T_a)_{oo} = 1$ for every a. All report transitions from o then return to o. Their probabilities, and the terminal readouts at o, constitute a one-label realization. In the case under consideration this is excluded. Hence

$$\beta_j = \frac{\overline{T}_{oj}}{1 - \overline{T}_{oo}} \quad (j \neq o)$$

is a probability row and $f_o = \sum_{j\neq o} \beta_j f_j$.

Delete label o. On the remaining labels define

$$\widehat{T}_{a,y}(i,j) = T_{a,y}(i,j) + T_{a,y}(i,o)\beta_j, \qquad i, j \neq o, \tag{9.5}$$

and retain their terminal decoders. These matrices are nonnegative and their sums over y are stochastic. The recursion $f_i((a,y)w) = \sum_j T_{a,y}(i,j) f_j(w)$, with the functional identity for f_o substituted on its right side, is exactly the recursion for (9.5). Induction on word length, starting with the empty word and each terminal decoder, proves that every retained f_i is unchanged. Initial

row β therefore realizes f_o with one fewer label. This proves $n_{\mathrm{mix}} \le n_{\mathrm{pure}} - 1$ whenever a finite pure realization exists.

Conversely, given a k-label mixed-start realization $(\mu, T_{a,y})$, append one label o. Its report transition on command a enters the core with row $\mu T_{a,y}$ and never returns to o; the core retains its original transitions. A terminal readout at o has row $\mu d_{b,z}$. The first physical command is already included in this row, so induction gives the original law at every word length, including zero. This uses $k+1$ labels and proves the reverse inequality. A finite pure realization is itself a finite mixed one, so the two finiteness assertions are equivalent. The one-label exception is immediate. □

Corollary 9.2 (Stationary processes)**.** *For every stationary finite-alphabet process that is not i.i.d., the minimum pure-start edge-emitting hidden Markov realization has exactly one more label than the minimum freely mixed realization, with the same infinite-value convention. In particular, the q versus $q+1$ conclusion of Theorem S14.1 is an instance of a general stationary principle, while its phase-dependent q remains a separate spectral conclusion.*

Equality of these complete response functions also preserves the law under every common randomized policy for commands, stopping and terminal readout. For each finite stopped history, multiply the equal response by the same policy probabilities. Summation gives all finite stopping events, and decreasing continuation events give the nontermination probability. Consequently all nonnegative costs measurable from that record have equal expectations, including infinite expectations. These are process-realization label counts. If the external policy has its own persistent memory, that memory is counted separately in a closed-loop implementation; it is not supplied free by the theorem.

10. The causal task-order transfer theorem

Suppose that every terminating state-stopped competitor reaches at least $B_0 \ge 1$ live labels under θ_*. Such a certificate must concern actual physical command compatibility, not printed action names. Define

i	$m_-(M)$	$m_+(M)$
cp	M	$\lfloor M/D_0 \rfloor$
ev	M	$\lfloor M/D_0 \rfloor$
st	$M - B_0$	$\lfloor (M-B)/D_0 \rfloor$.

(10.1)

The event upper also requires $M \ge B$. The trivial state-stopped lower $B_0 = 1$ is valid for a procedure initially waiting to make a call.

Theorem 10.1 (Affine task-order transfer with a paid reference and common construction)**.** *Assume all the prepared-kernel, full-report, task-order, effort, occupation, resource and common-construction hypotheses of Section 2. Fix N, M, put $b_- = N/\kappa_-$, $b_+ = N/\kappa_+$, and suppose $b_- \ge 1$ and $m_-(M) \ge 1$. Set*

$$t_N = (1 - A_N)_+ / u_-.$$

If $t_N > 0$, every one of the three risks satisfies

$$R_i^{\Theta}(N, M) \ge \left[l_- t_N \mathcal{Q}_{m_-(M)}(b_-/t_N) - E_N \right]_+. \tag{10.2}$$

If $t_N = 0$, the asserted lower is zero, with no evaluation of $\mathcal{Q}$ at infinity. An empty legal class still has risk $+\infty$.

Suppose also $m_+(M) \ge 1$, $b_+ \ge 1$, and, for event stopping, $M \ge B$. If

$$\beta_N = \frac{l_+ b_+}{1 + b_+ a_+} \ge 1, \tag{10.3}$$

one common Borel program with actual expected cost at most N in every model attains

$$R_i^{\Theta}(N, M) \le \frac{u_+(1 + b_+ a_+)}{l_+} \mathcal{Q}_{m_+(M)}(\beta_N) + b_+ e_+. \tag{10.4}$$

It commits almost surely. If $\beta_N < 1$, the always-accept first-attempt construction instead gives $R_i^{\Theta}(N,M) \le u_+ \mathcal{Q}_{m_+(M)}(1) + e_+$ at cost at most $\kappa_+ \le N$.

The lower is against every legal competing strategy and uses one actual witness. The upper is one parameter-independent strategy for all models. The two effort constants need not agree. Neither direction assumes that the comparison experiment belongs to Θ or that a normalized predictive update is uniformly contracting.

Corollary 10.2 (Scale-uniform relative transfer)**.** *Under Theorem 10.1, suppose $A_N = E_N = a_+ = e_+ = 0$. If $l_+ b_+ \ge 1$, then*

$$\frac{l_-}{u_-}\mathcal{Q}_{m_-(M)}(u_- b_-) \le R_i^{\Theta}(N,M) \le \frac{u_+}{l_+}\mathcal{Q}_{m_+(M)}(l_+ b_+). \tag{10.5}$$

The factors $l_\pm, u_\pm$ may be fixed nonunit constants, independent of budget and resolution. In particular no term proportional to $b_\pm$ times a fixed absolute calibration error is required in this relative-order class.

Proof. Substitute $t_N = 1/u_-$ and $\beta_N = l_+ b_+$ in the theorem. □

Corollary 10.3 (Separate exact checkpoint and autonomous laws)**.** *Suppose the certificates are exact: $l_\pm = u_\pm = 1$, all defects vanish, $\kappa_- = \kappa_+ = \kappa$, and exports are the identity. Then, for $N \ge \kappa$,*

$$R_{\mathrm{cp}}^{\Theta}(N,m) = \mathcal{Q}_m(N/\kappa).$$

The constructed autonomous upper uses $B + m$ state-stopped labels, or $\max(B,m)$ event-stopped labels. If the mandatory live count is sharp, $B_0 = B$, then

$$R_{\mathrm{st}}^{\Theta}(N, B+m) = \mathcal{Q}_m(N/\kappa) \quad (m \ge 1), \tag{10.6}$$

$$R_{\mathrm{ev}}^{\Theta}(N,M) = \mathcal{Q}_M(N/\kappa) \quad (M \ge B). \tag{10.7}$$

The subtraction by B belongs only to the state-stopped model. Nonidentity exports require the expanded dictionary in (10.1)*, unless an optimal dictionary is invariant under all exports. These exact stopping distinctions are inherited from the complete companion and are reproved here to fix the resource semantics.*

Proof. The same lower and upper profile coincide in each asserted interface. Identity exports give $D_0 = 1$. The packing and converse widths are proved in Lemma 11.4. No lower on live states is used to forbid event-stopped reuse. □

The familiar absolute-defect theorem is also a specialization: take $l_\pm = u_\pm = 1$ while retaining A_N, E_N, a_+, e_+. Thus relative and absolute calibration are composed in one statement. The chosen metric is the common scored task throughout; it is not changed between the converse and the construction.

11. Proof of the central theorem

Lemma 11.1 (Profile attainment)**.** *For every finite $m \ge 1$ and $b \ge 1$, the minimum in* (2.2) *is attained. The profile is nonincreasing in m, b and left-continuous in $b > 1$.*

Proof. Write a dictionary as an ordered tuple in the compact set H^m, permitting repetitions. The feasible densities are a weak-star closed subset of the radius-b ball of $L^\infty(\nu)$: positivity, the upper bound and the mass equality are each closed under pairing with $L^1(\nu)$. They are therefore weak-star compact. As the tuple varies, its squared-distance function varies uniformly on H, since all points belong to a bounded set. Along a convergent product net, split the objective difference into this uniform variation and the weak-star pairing against the fixed bounded measurable loss at the limit tuple. This proves continuity of the objective and attainment. Nets avoid an unwarranted sequential compactness assumption. Monotonicity follows from inclusion of feasible sets. For left continuity

take an optimizer h at $b > 1$ and let $h_\lambda = (1-\lambda)h+\lambda$. It has unit mass and cap $b_\lambda = (1-\lambda)b+\lambda < b$. Its objective tends to the optimum as $\lambda \downarrow 0$, while monotonicity bounds the optimum at b_λ between the optimum at b and this objective. □

Lemma 11.2 (Actual-history comparison measure)**.** *Fix any robustly admissible competitor of cost at most N, with its fixed terminal dictionary C. There are finite nonnegative measures ξ_c on X, $c \in C$, such that $\xi = \sum_c \xi_c$ obeys*

$$\xi \le (\mathbb{E}_* A_\tau)\nu \le b_-\nu, \tag{11.1}$$

$$1 \le u_-\xi(X) + A_N, \tag{11.2}$$

$$\mathbb{E}_*\|Z - \widehat{r}\|^2 \ge l_- \sum_{c\in C} \int d_c \, d\xi_c - E_N. \tag{11.3}$$

The measure ξ is not asserted to be the terminal law of any comparison execution.

Proof. Adjoin an independent uniform random variable to the current report on both sides. This is current-call randomness, not a seed retained between calls; it is sufficient to represent the competitor's randomized output and stopping decision. A finite-output kernel has such a representation by cumulative probabilities. At a fixed reached productive history h and chosen command a, the post-report decision then gives an acceptance indicator $q_h(y)$ and an output $r_h(y) \in C$ on acceptance. Give r_h an arbitrary fixed value on rejection. The notation y now includes the current independent random variable. Integrating the original certificates over that variable preserves their constants. Dependence on the entire earlier history is permitted in these functions.

For each Borel $D \subset X$, set

$$\xi_c(D) = \mathbb{E}_* \sum_{j\le\tau:\ j\ mproductive} \int \mathbf{1}_{\{\pi_{H_{j-1},a_j}(y)\in D\}} q_{H_{j-1}}(y) \mathbf{1}_{\{r_{H_{j-1}}(y)=c\}} P^0_{H_{j-1},a_j}(dy, H). \tag{11.4}$$

The outer expectation is the actual pre-call occupation at θ_*. All terms are jointly measurable and nonnegative. Tonelli defines the measures, whose total is finite because $\mathbb{E}_* A_\tau \le b_-$. Summing over c and dropping the acceptance indicator gives (11.1): each comparison row has mark law ν.

The analogous sum with the actual rows instead of P^0 has mass exactly one, because the actual procedure commits exactly once almost surely. Apply (2.4) at each actual reached history, then sum. The occupation certificate yields (11.2). Apply (2.5) in the same way to the disjoint actual commitment events. Their expected losses sum to the actual risk. The comparison contribution is $d_{r_h(y)}(\pi_{h,a}y)$ by (2.3); this identity conditions on the full current report and therefore does not overlook extra information seen by the competitor. The accumulated loss defect is at most E_N, giving (11.3). One may first truncate the number of calls. Monotone convergence for the nonnegative contributions and the finite defect bounds justify the limit.

No actual past is replaced by a comparison past. In particular the competitor need not terminate, or obey a cost bound, if hypothetically run under the comparison experiment. This distinction is why the argument applies when that experiment is not an operating model. □

Remark 11.3 (Why the full report is required). A constant mark can conceal a visible fair bit which also equals the future audit. A comparison using only that mark has $f = 0, w = 1$, but a two-value decoder reading the bit predicts it with zero loss. The hypotheses therefore condition on the full report seen by the competitor; matching a compressed mark marginal alone cannot establish the lower.

Lemma 11.4 (Live roles, output roles and physical compatibility)**.** *A program with ℓ live roles and t output roles has a state-stopped implementation with $\ell+t$ labels and an event-stopped implementation with $\max(\ell, t)$ labels, preserving every path cost and complete stopped scored law. Conversely an event-stopped M-label program has at most M decoder values. A state-stopped M-label program with at least B_0 reached live labels has at most $M - B_0$ terminal decoder values.*

For a physical live-state certificate, consider finitely many mandatory pre-call contexts I with no free context input. Suppose the commands legal at a history of context i are exactly $\mathcal{U}_i$, and every terminating competitor visits every context with positive probability under the same witness. This positive probability may depend on the competitor; no uniform lower bound over competitors is required. Let b_{phys} be the least size of a cover of I by subsets J with $\bigcap_{i\in J}\mathcal{U}_i \neq \varnothing$. Then at least b_{phys} live labels are required. This certificate is invariant under duplicate names and bijective stateless recodings of the same physical commands.

Proof. For state stopping take disjoint copies of the live and terminal role sets and type them zero and one. For event stopping embed the live and terminal role sets separately into one alphabet of size $\max(\ell, t)$. On continuing transitions use the live embedding; on commitment emit the final event and use the terminal embedding. A command row is read only on a continuing branch, while the decoder is read only after commitment. Sharing their label therefore causes no conflict and uses no continuing mode bit. Couple the original and recoded coins. Every finite stopped prefix and every path cost agree, and almost-sure termination gives equality of the stopped scored laws. A fixed decoder has one value per retained label. In state stopping the global halt map excludes each reached live label from the output set, proving its subtraction. Event stopping has no such exclusion.

A live label reached in contexts J must have one command distribution supported on all the corresponding $\mathcal{U}_i$. Since J is finite, the union of the exceptional null command sets is null; hence their intersection supports a probability and is nonempty. The sets of contexts served by the labels cover I, proving the bound. Conversely a nonempty intersection contains a command and hence a Dirac command distribution. Replacing names without changing executed kernels, legal domains, costs and reports does not change these intersections. A translator which itself remembers a hidden phase is not such a recoding; its memory must be charged. □

Proof of Theorem 10.1. Fix a competing program. Lemma 11.4 bounds its decoder width by $m_-(M)$. Let $s = \xi(X)$ in Lemma 11.2. If $t_N > 0$, then (11.2) gives $s \geq t_N$. Scale the measure to $\xi' = (t_N/s)\xi$. It has mass t_N and remains bounded above by $b_-\nu$. Thus $h = (d\xi'/d\nu)/t_N$ has unit mass and cap b_-/t_N. Nearest-dictionary projection and nonnegativity imply

$$\sum_{c\in C}\int d_c\, d\xi_c \geq \int [w + \operatorname{dist}(f, C)^2]\, d\xi \geq t_N\, \mathcal{Q}_{m_-(M)}(b_-/t_N).$$

Together with (11.3) this proves the asserted lower for this competitor. Its worst-model risk dominates its risk at θ_*. Taking the infimum over competitors gives (10.2), without exchanging an infimum and a supremum. Nonnegative loss gives the zero truncation and the stated case $t_N = 0$.

For the upper assume (10.3). Choose an attained optimizer (C, h) at width $m_+(M)$ and cap β_N. Let $r_C(x)$ be the first nearest point to $f(x)$ in a fixed ordering of C. After each attempt of $\mathcal{G}$, accept x with probability $h(x)/\beta_N$, export $T_J r_C(x)$ and commit; otherwise return to the start routing label. All these operations are Borel. The choice J is used within the current attempt; any part retained thereafter is charged through the live role or exported decoder, not a free seed. The exports are assumed jointly measurable and take at most $D_0|C|$ distinct values, so their finite decoder and transitions are Borel as well.

Conditional on every surviving attempt history, the acceptance probability is at least

$$p = \frac{l_+}{\beta_N} - a_+ = \frac{1}{b_+} > 0. \tag{11.5}$$

Consequently the probability of reaching attempt $j+1$ is at most $(1-p)^j$, by induction on conditional survival probabilities. In particular commitment is almost sure and the expected number of attempts is at most $1/p = b_+$. Each begun attempt has conditional expected cost at most κ_+, not merely

a cost conditional on a favorable outcome. Summing the predictable attempt costs gives actual expected cost at most $\kappa_+ b_+ = N$.

The conditional unnormalized loss contributed by an attempt is at most $u_+ \mathcal{Q}_{m_+(M)}(\beta_N)/\beta_N + e_+$. Sum its contributions on the disjoint terminal events, using the same survival bound. The result is at most

$$b_+ \left[\frac{u_+}{\beta_N} \mathcal{Q}_{m_+(M)}(\beta_N) + e_+ \right],$$

which is (10.4). The program has B live roles and at most $D_0 m_+(M)$ terminal roles. Lemma 11.4 gives exactly the sufficient widths in (10.1). A checkpoint counts only the terminal width. No physical renewal law was used: the conditional hazard controls survival even if the hidden state after rejection differs arbitrarily from its former state.

If $\beta_N < 1$, commit after the first attempt, and use an optimizer at cap one. The loss certificate with $q = 1$ gives $u_+ \mathcal{Q}_{m_+(M)}(1) + e_+$, and the whole-attempt cost is at most κ_+. This proves all remaining assertions. □

12. Scope and further mathematical questions

The shared-preparation normal form applies to every finite group, every finite preparation and outcome alphabet, and closed convex wordwise specifications. The input controller need not satisfy group relations. The invariant polytope may extend beyond the target response span. The exact low-contrast formula requires a common mean law; the displayed nonnegative-rank counterexample explains why this hypothesis cannot simply be removed. These are shared-device results with one fixed decoder, not a claim that independently optimized devices can always share their memory at no cost.

The reset classification allows multiple commands and tied optimal actions, but uses positive failure effects and the finite reset property. The explicit positive-semidefinite counterexample shows the role of reachability under an exposing preparation. The finite gap certificate detects failure to attain the unrestricted optimum; it is not the full memory-constrained cost frontier. The complexity reduction supplies a concrete limitation on uniform efficient synthesis, not an impossibility of useful algorithms for particular experimental families.

The new approximation lower permits arbitrary classical terminal decoder tables but requires delayed terminal test requests and no uncounted access to the preparation history. At fixed positive target budget and the original algebraic phase it leaves a gap between logarithmic-in-logarithm and logarithmic memory growth. The sharper matching logarithmic physical-dictionary law retains its Born-readout and badly-approximable hypotheses. Neither law is the original general-M squared-sign minimax-risk frontier. The latter, general nonrigid optimal feedback without the reset/reachability hypotheses, and the intermediate phase-type timing frontier remain distinct mathematical problems.

The inherited mean–contrast law is exact for its displayed growing symmetric-group family, and the general finite-group error principle is an attained finite optimization. No closed formula for every group or universal polynomial-time synthesis is inferred. The return-polytope construction concerns explicitly verified conditional branches, not average mixing. Its target-cost bound does not control the simulator's unbounded expected duration. The stationary-initialization theorem concerns exact predictively stationary laws; an approximate truncation does not automatically inherit its extra-label lower bound.

All original causal, predictive-state, information-compression, finite resource, finite-order measurement, two-stage reset, rank-constrained discrimination, communication, dual, spectral and geometric developments retain their hypotheses. Local geometry remains local. Independent A1, A2 and further Theta construction problems require their own arguments. Program bits, arithmetic precision, working storage and sampling work are not silently identified with persistent label counts. The paired supplement contains complete technical and independent proofs; the separately compilable R58 companion retains its own scope.

References

[1] D. Blackwell, *Equivalent comparisons of experiments*, Ann. Math. Statist. **24** (1953), 265–272. DOI: 10.1214/aoms/1177729032.

[2] J. A. Cuesta-Albertos, A. Gordaliza and C. Matrán, *Trimmed k-means: an attempt to robustify quantizers*, Ann. Statist. **25** (1997), 553–576. DOI: 10.1214/aos/1031833664.

[3] L. A. García-Escudero and A. Gordaliza, *Robustness properties of k means and trimmed k means*, J. Amer. Statist. Assoc. **94** (1999), 956–969. DOI: 10.1080/01621459.1999.10474200.

[4] C. R. Shalizi and J. P. Crutchfield, *Computational mechanics: pattern and prediction, structure and simplicity*, J. Statist. Phys. **104** (2001), 817–879. DOI: 10.1023/A:1010388907793.

[5] N. Barnett and J. P. Crutchfield, *Computational mechanics of input–output processes: structured transformations and the ε-transducer*, J. Statist. Phys. **161** (2015), 404–451; arXiv:1412.2690.

[6] A. Ambainis, M. Mosca, A. Tapp and R. de Wolf, *Private quantum channels*, Proc. 41st Annual Symposium on Foundations of Computer Science (2000), 547–553. DOI: 10.1109/SFCS.2000.892142.

[7] F. Buscemi, *Comparison of quantum statistical models: equivalent conditions for sufficiency*, Comm. Math. Phys. **310** (2012), 625–647. DOI: 10.1007/s00220-012-1421-3.

[8] N. Bertschinger and J. Rauh, *The Blackwell relation defines no lattice*, Proc. IEEE International Symposium on Information Theory (2014), 2479–2483. DOI: 10.1109/ISIT.2014.6875280; arXiv:1401.3146.

[9] A. Banerjee, X. Guo and H. Wang, *On the optimality of conditional expectation as a Bregman predictor*, IEEE Trans. Inform. Theory **51** (2005), 2664–2669. DOI: 10.1109/TIT.2005.850145.

[10] B. von Stengel, *Efficient computation of behavior strategies*, Games Econom. Behav. **14** (1996), 220–246. DOI: 10.1006/game.1996.0050.

[11] E. J. Balder, *On compactness of the space of policies in stochastic dynamic programming*, Stochastic Process. Appl. **32** (1989), 141–150. DOI: 10.1016/0304-4149(89)90058-6.

[12] J. Barry, D. T. Barry and S. Aaronson, *Quantum partially observable Markov decision processes*, Phys. Rev. A **90** (2014), 032311. DOI: 10.1103/PhysRevA.90.032311; arXiv:1406.2858.

[13] S. Ying and M. Ying, *Reachability analysis of quantum Markov decision processes*, Inform. Comput. **263** (2018), 31–51. DOI: 10.1016/j.ic.2018.09.001; arXiv:1406.6146.

[14] N. Saldi, S. Sanjari and S. Yüksel, *Quantum Markov decision processes: general theory, approximations, and classes of policies*, SIAM J. Control Optim. **63** (2025), 1156–1183. DOI: 10.1137/24M1643864; arXiv:2402.14649.

[15] A. Seidenberg, *A new decision method for elementary algebra*, Ann. of Math. (2) **60** (1954), 365–374. DOI: 10.2307/1969640.

[16] K. Chatterjee, M. Chmelík, R. Gupta and A. Kanodia, *Optimal cost almost-sure reachability in POMDPs*, Proc. AAAI Conference on Artificial Intelligence **29** (2015). DOI: 10.1609/aaai.v29i1.9683.

[17] R. Andriushchenko, A. Bork, M. Češka, S. Junges, J.-P. Katoen and M. Macák, *Search and explore: symbiotic policy synthesis in POMDPs*, Computer Aided Verification (2023); arXiv:2305.14149v2.

[18] P. R. Halmos, *Two subspaces*, Trans. Amer. Math. Soc. **144** (1969), 381–389. DOI: 10.1090/S0002-9947-1969-0251519-5.

[19] D. A. Kessler, E. Barkai and K. Ziegler, *First-detection time of a quantum state under random probing*, Phys. Rev. A **103** (2021), 022222. DOI: 10.1103/PhysRevA.103.022222; arXiv:2012.01763.

[20] G. Del Vecchio Del Vecchio and S. N. Majumdar, *Optimal detection of quantum states via projective measurements*, J. Phys. A: Math. Theor. **59** (2026), 035001. DOI: 10.1088/1751-8121/ae34fe; arXiv:2509.08556.

[21] Q. Qi, *General Theta Foundations I: Acquired Geometry and Causal Resource Transfer*, complete technical companion on absolute calibration, exact stopping resources and nonintervening sensing, 2026. The complete retained R58 native source and its independently built companion accompany this manuscript; their identities are fixed in the source-provenance and preservation records.

[22] W.-G. Tzeng, *A polynomial-time algorithm for the equivalence of probabilistic automata*, SIAM J. Comput. **21** (1992), 216–227. DOI: 10.1137/0221017.

[23] Y. Li, N. Yu and M. Ying, *Termination of nondeterministic quantum programs*, arXiv:1201.0891 (2012).

[24] J. Fu, H. Jiang, M. Xu, Y. Deng and Z.-B. Li, *Algorithmic analysis of termination problems for nondeterministic quantum programs*, arXiv:2402.15827 (2024).

[25] J. Liu and S. Verdú, *Rejection sampling and noncausal sampling under moment constraints*, Proc. IEEE International Symposium on Information Theory (2018), 1565–1569. DOI: 10.1109/ISIT.2018.8437857.

[26] D. Goc and G. Flamich, *On channel simulation with causal rejection samplers*, Proc. IEEE International Symposium on Information Theory (2024). DOI: 10.1109/ISIT57864.2024.10619339; examined text arXiv:2401.16579v2 (3 May 2024).

[27] Z. Li and T. Liu, *Ranking experiments under sequential sampling*, arXiv:2608.19897v2 (28 August 2026). Version 1 appeared under the title *The order of binary experiments under endogenous stopping*, with Z. Li as sole author; the version-2 statement is the one compared here.

[28] S. Ying and M. Ying, *Reachability analysis of quantum Markov decision processes*, arXiv:1406.6146 (2014), Theorem 3.6; published journal version listed above as [13].
[29] A. D. Kara and S. Yüksel, *Robustness to incorrect system models in stochastic control*, SIAM J. Control Optim. **58** (2020), 1144–1182. DOI: 10.1137/18M1208058; examined text arXiv:1803.06046v7.
[30] Y. E. Demirci, A. D. Kara and S. Yüksel, *Sensitivity of filter kernels and robustness bounds to transition and measurement kernel perturbations in partially observable stochastic control*, arXiv:2508.10658v3 (2025).
[31] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018, Chapter 3, Corollary 3.40.
[32] K. Kuratowski and C. Ryll-Nardzewski, *A general theorem on selectors*, Bull. Acad. Polon. Sci. Sér. Sci. Math. Astronom. Phys. **13** (1965), 397–403.
[33] A. Charnes, W. W. Cooper and K. O. Kortanek, *Duality in semi-infinite programs and some works of Haar and Carathéodory*, Management Sci. **9** (1963), 209–228. DOI: 10.1287/mnsc.9.2.209.
[34] A. Shapiro, *Semi-infinite programming, duality, discretization and optimality conditions*, Optimization **58** (2009), 133–161. DOI: 10.1080/02331930902730070; author preprint, Optimization Online (2008).
[35] F. Dufour and T. Prieto-Rumeau, *Absorbing Markov decision processes and their occupation measures*, SIAM J. Control Optim. **63** (2025), 676–698. DOI: 10.1137/24M1661959. Related author preprint examined: *A note on weak compactness of occupation measures for an absorbing Markov decision process*, arXiv:2402.10672v2 (2024).
[36] V. Franc, D. Prusa and V. Voracek, *Optimal strategies for reject option classifiers*, J. Mach. Learn. Res. **24** (2023), no. 11, 1–49.
[37] A. Jenčová, *Comparison of quantum channels and statistical experiments*, arXiv:1512.07016v2 (2016). The separately indexed shorter version arXiv:1601.06370 was withdrawn in favor of this extended text.
[38] C. T. Li and A. El Gamal, *Strong functional representation lemma and applications to coding theorems*, IEEE Trans. Inform. Theory **64** (2018), 6967–6978; arXiv:1701.02827v4.
[39] D. Aldous and L. Shepp, *The least variable phase type distribution is Erlang*, Comm. Statist. Stochastic Models **3** (1987), 467–473. DOI: 10.1080/15326348708807067.
[40] M. Sion, *On general minimax theorems*, Pacific J. Math. **8** (1958), 171–176. DOI: 10.2140/pjm.1958.8.171.
[41] A. Piunovskiy and Y. Zhang, *Extreme occupation measures in Markov decision processes with an absorbing state*, SIAM J. Control Optim. **62** (2024), 65–90. DOI: 10.1137/23M1572398. Author text examined: *Extreme occupation measures in Markov decision processes with a cemetery*, arXiv:2307.03158v1.
[42] F. Dufour and T. Prieto-Rumeau, *Absorbing Markov decision processes: geometric properties and sufficiency of finite mixtures of deterministic policies*, arXiv:2512.17511; examined HTML at version-1 URL, Theorems 4.3–4.4 and Proposition 4.5.
[43] C. Mereghetti, B. Palano and P. Raucci, *Unary quantum finite state automata with control language*, Appl. Sci. **14** (2024), no. 4, 1490. DOI: 10.3390/app14041490.
[44] M. O. Rabin and D. Scott, *Finite automata and their decision problems*, IBM J. Res. Develop. **3** (1959), 114–125. DOI: 10.1147/rd.32.0114.
[45] H. Jaeger, *Observable operator models for discrete stochastic time series*, Neural Comput. **12** (2000), 1371–1398. DOI: 10.1162/089976600300015411.
[46] M. Thon and H. Jaeger, *Links between multiplicity automata, observable operator models and predictive state representations—a unified learning framework*, J. Mach. Learn. Res. **16** (2015), 103–147.
[47] A. Monras and A. Winter, *Quantum learning of classical stochastic processes: the completely-positive realization problem*, LIPIcs **27**, TQC 2014 (2014), 99–109. DOI: 10.4230/LIPIcs.TQC.2014.99.
[48] A. Heller, *On stochastic processes derived from Markov chains*, Ann. Math. Statist. **36** (1965), 1286–1291. DOI: 10.1214/aoms/1177700000.
[49] J. W. Carlyle and A. Paz, *Realizations by stochastic finite automata*, J. Comput. System Sci. **5** (1971), 26–40. DOI: 10.1016/S0022-0000(71)80005-3.
[50] M. Fanizza, J. Lumbreras and A. Winter, *Quantum theory in finite dimension cannot explain every general process with finite memory*, Comm. Math. Phys. **405** (2024), article 50. DOI: 10.1007/s00220-023-04913-4; arXiv:2209.11225. See Section 2.3 for the FRDN phase examples and Theorem 2 and Section 4.2 for the noise-dependent worst-phase classical-memory lower bound.
[51] N. Dmitriev and E. Dynkin, *On characteristic roots of stochastic matrices*, Izv. Akad. Nauk SSSR Ser. Mat. **10** (1946), no. 2, 167–184. Math-Net.Ru: im3595 (Russian).
[52] C. R. Johnson and P. Paparella, *A matricial view of the Karpelevič theorem*, Linear Algebra Appl. **520** (2017), 1–15. DOI: 10.1016/j.laa.2017.01.009; arXiv:1611.06970v2.
[53] S. Kirkland and H. Šmigoc, *Stochastic matrices realising the boundary of the Karpelevič region*, Linear Algebra Appl. **635** (2022), 116–138. DOI: 10.1016/j.laa.2021.11.016; arXiv:2110.01040v1.
[54] L. Benvenuti and L. Farina, *A tutorial on the positive realization problem*, IEEE Trans. Automat. Control **49** (2004), 651–664. DOI: 10.1109/TAC.2004.826715.
[55] L. Benvenuti and L. Farina, *An example of how positivity may force realizations of "large" dimension*, Systems Control Lett. **36** (1999), 261–266. DOI: 10.1016/S0167-6911(98)00098-X.

[56] C. A. O'Cinneide, *Characterization of phase-type distributions*, Comm. Statist. Stochastic Models **6** (1990), 1–57. DOI: 10.1080/15326349908807134.

[57] R. S. Maier, *The algebraic construction of phase-type distributions*, Comm. Statist. Stochastic Models **7** (1991), 573–602. DOI: 10.1080/15326349108807207.

[58] A. N. W. Hone and J. L. Varona, *Continued fractions and irrationality exponents for modified Engel and Pierce series*, Monatsh. Math. **190** (2019), 501–516. DOI: 10.1007/s00605-018-1244-1. Author version arXiv:1810.10127v1, equation (3.3).

[59] R. D. Nussbaum and M. Scheutzow, *Admissible arrays and a nonlinear generalization of Perron–Frobenius theory*, J. London Math. Soc. (2) **58** (1998), 526–544. DOI: 10.1112/S0024610798006796.

[60] D. R. Brown, On clans of non-negative matrices, *Proc. Amer. Math. Soc.* **15** (1964), 671–674, doi:10.1090/S0002-9939-1964-0165504-6.

[61] Š. Schwarz, On the structure of the semigroup of stochastic matrices, *Publ. Math. Inst. Hungar. Acad. Sci.* **9** (1964), 297–311, especially Theorems 7–8.

[62] P. Flor, On groups of non-negative matrices, *Compositio Math.* **21** (1969), 376–382, `https://www.numdam.org/article/CM_1969__21_4_376_0.pdf`.

[63] J.-P. Serre, Bounds for the orders of the finite subgroups of $G(k)$, in *Group Representation Theory*, M. Geck, D. Testerman and J. Thévenaz (eds.), EPFL Press, Lausanne, 2007, 405–450; author lecture text, April 2006, archived as arXiv:1011.0346.

[64] L. Babai, R. Beals and D. Rockmore, Deciding finiteness of matrix groups in deterministic polynomial time, *Proceedings of ISSAC 1993*, 117–126, doi:10.1145/164081.164104.

[65] S. Kiefer and B. Wachter, Stability and complexity of minimising probabilistic automata, in *ICALP 2014, Part II*, Lecture Notes in Computer Science **8573**, 268–279, doi:10.1007/978-3-662-43951-7_23; arXiv:1404.6673, Section 4.2.

[66] Š. Schwarz, Homomorphisms of a completely simple semigroup onto a group, *Matematicko-fyzikálny časopis* **12** (1962), no. 4, 293–300; `https://www.dml.cz/handle/10338.dmlcz/126333`.

[67] J. Gouveia, P. A. Parrilo and R. R. Thomas, *Lifts of convex sets and cone factorizations*, Math. Oper. Res. **38** (2013), 248–264. DOI: 10.1287/moor.1120.0575; arXiv:1111.3164, especially Theorem 2.12 on equivariant factorizations.

[68] V. Kaibel, K. Pashkovich and D. O. Theis, *Symmetry matters for sizes of extended formulations*, SIAM J. Discrete Math. **26** (2012), 1361–1382. DOI: 10.1137/110839813; arXiv:0911.3712v3. An extended abstract appeared in IPCO 2010, LNCS 6080, 135–148.

[69] C. Bocci, E. Carlini and F. Rapallo, *Perturbation of matrices and nonnegative rank with a view toward statistical models*, SIAM J. Matrix Anal. Appl. **32** (2011), 1500–1512. DOI: 10.1137/110825455; arXiv:1010.3566.

[70] N. Gillis and F. Glineur, *On the geometric interpretation of the nonnegative rank*, Linear Algebra Appl. **437** (2012), 2685–2712. DOI: 10.1016/j.laa.2012.06.038; arXiv:1009.0880.

[71] C. Okay, *Polyhedral classical simulators for quantum computation*, Res. Math. Sci. **13** (2026), article 55. DOI: 10.1007/s40687-026-00626-x. In particular Theorem 16, Lemma 17 and Corollary 19 treat invariant-polytope updates and adaptive composition.

[72] G. Aubrun and S. J. Szarek, *Dvoretzky's theorem and the complexity of entanglement detection*, Discrete Anal. (2017), paper no. 1, 20 pp.; arXiv:1510.00578, especially Section 6 on polyhedral approximations of density matrices.

[73] H. Fawzi, *On polyhedral approximations of the positive semidefinite cone*, Math. Oper. Res. **46** (2021), 1479–1489. DOI: 10.1287/moor.2020.1077; arXiv:1811.09649.

[74] A. E. Holroyd and T. M. Liggett, *Finitely dependent coloring*, Forum Math. Pi **4** (2016), e9, 43 pp. DOI: 10.1017/fmp.2016.7; arXiv:1403.2448. See Section 5 and the finite hidden-Markov obstruction attributed there to Schramm.

[75] J.-P. Serre, *Linear Representations of Finite Groups*, Graduate Texts in Mathematics **42**, Springer-Verlag, New York, 1977. DOI: 10.1007/978-1-4684-9458-7.

[76] A. Monras and A. Winter, *Quantum learning of classical stochastic processes: the completely-positive realization problem*, J. Math. Phys. **57** (2016), 015219. DOI: 10.1063/1.4936935; arXiv:1412.3634.

[77] O. Vu Thanh, N. Gillis and F. Lecron, *Bounded simplex-structured matrix factorization: algorithms, identifiability and applications*, IEEE Trans. Signal Process. **71** (2023), 2434–2447. DOI: 10.1109/TSP.2023.3289704; arXiv:2209.12638v2, especially Section IV-C and Theorem 3.

[78] N. Gillis, *Sparse and unique nonnegative matrix factorization through data preprocessing*, J. Mach. Learn. Res. **13** (2012), 3349–3386, especially Section 4.3, Lemma 27, Example 2 and Theorem 29. `https://jmlr.org/papers/v13/gillis12a.html`.

[79] M. O. Rabin, *Probabilistic automata*, Information and Control **6** (1963), 230–245. DOI: 10.1016/S0019-9958(63)90290-0, especially Theorem 3 and Section VIII.

[80] S.-Y. R. Li, *A martingale approach to the study of occurrence of sequence patterns in repeated experiments*, Ann. Probab. **8** (1980), no. 6, 1171–1176, doi:10.1214/aop/1176994578.

[81] D. P. Bertsekas and J. N. Tsitsiklis, *An analysis of stochastic shortest path problems*, Math. Oper. Res. **16** (1991), 580–595. DOI: 10.1287/moor.16.3.580.

[82] N. Meuleau, K.-E. Kim, L. P. Kaelbling and A. R. Cassandra, *Solving POMDPs by searching the space of finite policies*, Proceedings of UAI (1999), 417–426; author's deposit arXiv:1301.6720.

[83] P. Poupart and C. Boutilier, *Bounded finite state controllers*, Advances in Neural Information Processing Systems **16** (2003).

[84] R. M. Karp, *Reducibility among combinatorial problems*, in R. E. Miller and J. W. Thatcher (eds.), Complexity of Computer Computations, Plenum Press (1972), 85–103. DOI: 10.1007/978-1-4684-2001-2_9.

[85] A. Vandaele, N. Gillis and F. Glineur, *On the linear extension complexity of regular n-gons*, Linear Algebra Appl. **521** (2017), 217–239. DOI: 10.1016/j.laa.2016.12.023; arXiv:1505.08031v2, especially Section 3.1 on support antichains and rectangle covering.